Adrenal Pathology



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Image: artsy interpretation of the adrenal glands secreting hormones (although it would do so by releasing them into the blood stream, not through paracrine release as is implied in this image)

LEARNING OBJECTIVES

By exam time, you should be able to do the following:

- Correlate the gross and histopathological features of a pheochromocytoma with its tissue of origin and its clinical manifestations.
- Describe and identify genetic syndromes commonly associated with pheochromocytomas.
- Correlate the gross and histopathological features of an aldosterone-producing adrenal cortical adenoma with the clinical findings of Conn syndrome.
- Correlate the gross and histopathological features of adrenal atrophy (for example, due to autoimmune adrenalitis) with the clinical findings of Addison disease.
- Describe additional causes of acute adrenal insufficiency, including Waterhouse–Friderichsen syndrome, hypopituitarism, and steroid withdrawal.
- Correlate the gross and histopathological features of secondary adrenal hyperplasia with the clinical findings of Cushing syndrome or disease.

The Learning Objectives (LOs) are included here. Note that the order of the LOs does not match the order of the cases in the TBL; this is by design. It encourages you to think through each case, rather that simply matching it to its LO to determine the diagnosis of interest. Ideally, you'll be looking at these notes *after* the TBL has concluded, during review, so here is the association of each LO with the case that represents it:

• Correlate the gross and histopathological features of a pheochromocytoma with its tissue of origin and its clinical manifestations. **Case 3**
• Describe and identify genetic syndromes commonly associated with pheochromocytomas. **Case 3**
• Correlate the gross and histopathological features of an aldosterone-producing adrenal cortical adenoma with the clinical findings of Conn syndrome. **Case 1**
• Correlate the gross and histopathological features of adrenal atrophy (for example, due to autoimmune adrenalitis) with the clinical findings of Addison disease. **Case 2**
• Describe additional causes of acute adrenal insufficiency, including Waterhouse–Friderichsen syndrome, hypopituitarism, and steroid withdrawal. **Case 2**
• Correlate the gross and histopathological features of secondary adrenal hyperplasia with the clinical findings of Cushing syndrome (disease). **Case 4**

Case #1

CASE #1

A 45-year-old patient presents to the endocrinologist with complaints of uncontrolled hypertension and muscle weakness. They have a history of hypertension for the past five years, which has been difficult to manage despite being on multiple antihypertensive medications. Lately, they have noticed increased fatigue and generalized weakness, particularly in the legs. The patient denies changes in weight, appetite, or urinary symptoms.

On examination, the patient's blood pressure is elevated at 160/100 mmHg. The cardiac examination is unremarkable. There is no significant edema. Muscle testing reveals mild weakness in the proximal lower limbs.

Q1: Think back to the Renal Block—what are some secondary causes of refractory hypertension?

Vascular diseases: renal artery stenosis, fibromuscular dysplasia, coarctation
Primary renal: glomerular disease
Endocrine: Cushing syndrome, hyperaldosteronism, pheochromocytoma, hyperthyroidism, metabolic syndrome
Pulmonary: obstructive sleep apnea

Our first case entails a middle-aged patient with refractory hypertension. Most cases of hypertension (>90%) are essential (or primary) hypertension—elevated blood pressure with no discernible underlying cause. A minority, <10%, are due to secondary causes of hypertension. Hypertension that is refractory to medical treatment should prompt a search for secondary causes. This patient also complains of muscle weakness, which is also evoked on physical examination.

Secondary causes of hypertension may be due to vascular etiologies, endocrine etiologies, and several others. A few such causes are listed above on the slide. In this TBL, of course, we'll focus on the endocrine etiologies.

CASE #1

The patient is on a thiazide diuretic, a calcium-channel blocker, and an ACE inhibitor for blood pressure management. They also take potassium chloride for hypokalemia. Additional lab testing reveals the following:

Test	Value	Normal Range
Na	139	136–145 mEq/L
K	2.9 (**L**)	3.5–5.0 mEq/L
Cl	100	95–105 mEq/L
CO2	30 (**H**)	22–28 mEq/L
BUN	14	7–18 mg/dL
Cr	1.0	0.6–1.2 mg/dL
Glucose	96	70–110 mg/dL

Q2: How do you interpret the patient's lab tests? What follow-up tests might you consider?

There is hypokalemia (despite repletion with KCl) and alkalosis (elevated CO_2, which is primarily a measure of serum HCO_3^-). This is concerning for a disturbance in the aldosterone system; renin activity, aldosterone concentration, and the ratio between them would be useful next tests.

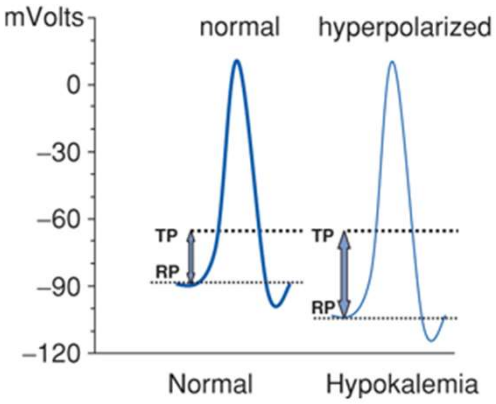
We get a bit more history, and it continues to suggest that there is a secondary cause for this patient's hypertension. They are on multiple medical therapies, and yet their blood pressure on examination had been 160/100 mmHg. Lab testing is notable for hypokalemia and, presumably, metabolic alkalosis (the elevated CO_2 really reflects an increase in serum bicarbonate concentration, which could be compensation for respiratory acidosis or primary metabolic alkalosis. This patient has no respiratory findings, so it unlikely an underlying respiratory cause). Putting these together, we may consider what would cause both potassium and hydrogen ion concentration to decrease (a decrease in $[H^+]$ would cause alkalosis). These two are linked through the function of **aldosterone**; aldosterone's primary function in the nephron is to cause retention of sodium (and, by extension, water) with the concomitant secretion of K^+ and H^+.

CASE #1

Q3: How might the abnormal lab tests be related to the muscle weakness the patient complained about, and seen on physical examination?

Hypokalemia may lead to muscle weakness by lowering the resting membrane potential to a more negative value. Remember, the dominant ion affecting the membrane potential is potassium itself, which has a higher concentration inside the cells than outside. Thus, lowering $[K^+]_{outside}$ increases the magnitude of the transmembrane gradient and increases the flux of potassium across the membrane. This drives the resting membrane potential toward the Nernst potential of potassium, which is –90 mV.

If that membrane physiology is throwing you for a loop, the takeaway is this: the membrane is hyperpolarized, making it harder for the cell to reach threshold and fire an action potential to elicit contraction. By extension, the muscle is weakened.



TP = threshold potential
RP = resting potential

This patient came in complaining of refractory hypertension but also had noted muscle weakness. Their hypokalemia may be to blame. Remember that potassium plays major roles in maintaining the resting membrane potential as well as in repolarization and hyperpolarization in the action potential. The normal gradient of potassium is such that the concentration inside of cells, $[K^+]_{inside}$, is higher than the concentration outside of them, $[K^+]_{outside}$. Thus, lowering of $[K^+]_{outside}$, as is the case in hypokalemia, will only increase that gradient. When the cell is at rest, its permeability is highest to potassium. Thus, with an increased gradient, there is increased flux of potassium at resting state. And that increased flux will pull the resting membrane potential toward the Nernst potential of K^+ at –90 mV (remember Goldman–Hodgkin–Katz, anyone?).

Enough basic physiology. Basically, when there is hypokalemia, the cell is sitting at a lower resting membrane potential, and that means the cell must be depolarized more to reach threshold. This decreases the likelihood of reaching that threshold and of eliciting an action potential, leading to weakness in the muscle.

CASE #1

The additional testing reveals the following:

Test	Value	Normal Range
Plasma renin activity	0.3	0.7–3.3 ng/mL/hr
Plasma aldosterone concentration	35	7–30 ng/dL
Aldosterone-to-renin ratio (ARR)	116	≤30 ng/dL per ng/mL/hr

Q4: How do these lab tests add to the interpretation of what is going on in the patient?

The patient's aldosterone concentration is high, but the renin activity is low. Accordingly, the aldosterone-to-renin ratio is markedly elevated. This must be a case of primary hyperaldosteronism (high aldosterone production on its own, rather than being stimulated by high renin levels). This is in contrast to secondary hyperaldosteronism, which would be driven by hyperreninemia (high renin).

Q5: What's the next step?

We must locate the source of aldosterone production via imaging (abdominal CT).

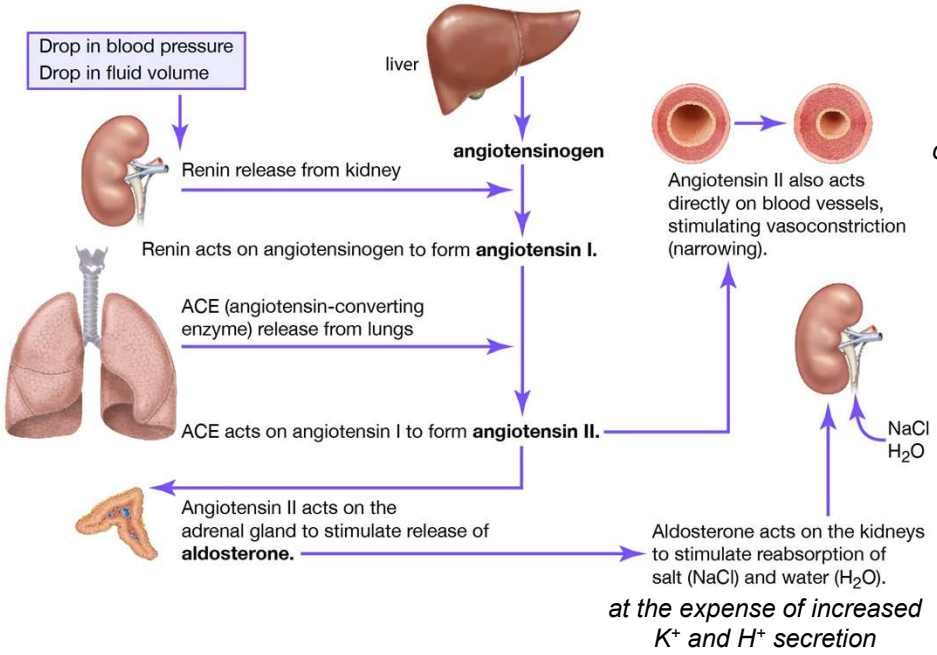
Let's try to make a diagnosis. When evaluating endocrine pathways, we'll just about always want to test more than hormone. Let's see why that is. We may be struck first by this patient's high **aldosterone** concentration (**hyperaldosteronism**). But is this a problem with the adrenal producing too much aldosterone on its own; that is, primary hyperaldosteronism? Or is it being told to do so by a **tropic hormone** (a hormone that signals for another endocrine tissue to perform its functions; note: this term tends to be used in the context of stimulating hormones from the anterior pituitary but would apply to any hormone acting on another endocrine tissue)—that is, secondary hyperaldosteronism? The major tropic hormone we're looking at here is **renin**, which should stimulate the **renin–angiotensin–aldosterone system**. But the **renin activity** is low. Thus, this must be a problem where aldosterone is being produced on its own, without being stimulated by renin. (In fact, the renin concentration is low because of negative feedback from the end-organ effects of aldosterone.) The **aldosterone-to-renin ratio** (**ARR**) is a simple measure calculated from the other two tests; values >30 ng/dL per ng/mL/hr is suggestive of primary hyperaldosteronism.

Now that we know that the issue is the high production of aldosterone, we should probably figure out where that aldosterone is coming from. Imaging could be helpful to do that, especially focusing on the adrenal gland (since it's the primary source of aldosterone, normally).

CASE #1

Before we see the CT, let's make sure we understand the physiology of the renin–angiotensin–aldosterone system.

Q6: Where do these hormones come from? How do they regulate each other? What are the end-organ effects of these hormones?



additional effects of angiotensin II:
release vasopressin
constrict efferent arterioles in kidney
increase sodium reabsorption
increase proton excretion
decrease medullary blood flow

Remember this old hormonal system from the Renal Block? It's back! Remember that the production and release **renin**, produced by the **juxtaglomerular cells** of the kidney, is stimulated by a decrease in renal perfusion. This is often due to a drop in blood pressure or fluid volume, although more specific causes, like renal artery stenosis, may also lead to decreased flow into the organ. Once in the circulation, renin can metabolize the conversion of **angiotensinogen** from the liver to **angiotensin I**. When angiotensin I reaches the lungs, it is acted upon by **angiotensin-converting enzyme** (**ACE**) to convert it to **angiotensin II**. Angiotensin II has its own functions, like causing vasoconstriction systemically and in the efferent arterioles of the kidney, and others as listed on the slide. But it also stimulates the adrenal gland to release **aldosterone**. Aldosterone then stimulates the kidney to increased its resorption of Na^+ while increasing its secretion of K^+ and H^+. The resorption of sodium should lead to increased resorption of water, thus increasing the blood pressure. And returning to the juxtaglomerular cells, this increased blood pressure should relieve the simulating that caused renin release in the first place; renin concentration should go back down to normal resting levels.

CASE #1

Q7: What syndrome does this patient have? What are the major etiologies of this syndrome that would be consistent with our patient's presentation?

Primary hyperaldosteronism may also be called Conn syndrome. There are two major causes: an aldosterone-producing adenoma (about one-third of cases) and bilateral adrenal hyperplasia without a discrete mass (two-thirds). There are some idiopathic cases that lack gross pathological changes to the adrenal glands, too.

Q8: What other lab testing might confirm the diagnosis?

Confirmatory Tests	Protocol	Results Suggestive of Primary Hyperaldosteronism	Rationale
Fludrocortisone suppression	Fludrocortisone acetate, 0.1 mg orally every 6 hours; sodium chloride, 30 mM every 8 hours; and potassium supplementation for 4 days	Upright PAC >5 ng/dL and PRA <1.0 ng/mL/h in the morning after 4 days	*fludrocortisone acts as a mineralocorticoid → should suppress native aldosterone production*
Intravenous saline load	2 L of 0.9% sodium chloride infusion over 4 hours	PAC after infusion >5 ng/dL	*give a lot of sodium → should suppress aldosterone production*
Oral saline load	300 mM sodium for 3 days, with potassium supplementation	Urinary aldosterone after 3 days >12 µg/24 h and urinary sodium >200 mM/24 h	

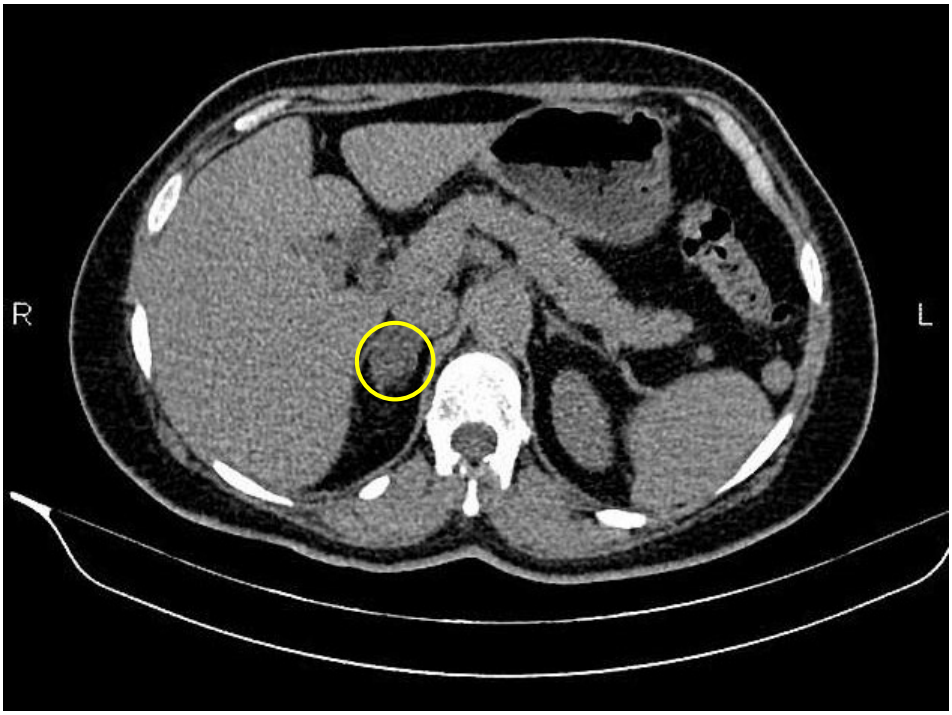
PAC = plasma aldosterone concentration; PRA = plasma renin activity

This patient's findings are consistent with **primary hyperaldosteronism** (**Conn syndrome**). There are two major causes of primary hyperaldosteronism. A notable minority of cases (about one-third) are due to a benign aldosterone-secreting tumor: an adrenal cortical adenoma (aldosteronoma). Most of the remaining two-thirds of cases are due to bilateral adrenal hyperplasia, without a distinct tumor. Some cases are idiopathic and don't fit into either category: the adrenal glands look grossly normal, but still produce excessive aldosterone.

As is often the case with endocrine pathologies, we may have a good screening test (the plasma aldosterone concentration, plasma renin activity, and aldosterone-to-renin ratio shown before) but may need a confirmation test to be definitive about the diagnosis. There are several options in this scenario; they are described in the slide. The takeaway for all of them is that we have a non-suppressible elevation in aldosterone. In each case, we set up a scenario that *should* cause aldosterone levels to fall. If they don't, that implies that there is pathology underlying the hyperaldosteronism. Shown here are the **fludrocortisone suppression test**, in which fludrocortisone (a mineralocorticoid) is given. This should lead to similar effects to the aldosterone itself and should, though negative feedback, suppress the patient's endogenous aldosterone production. Failure for aldosterone to fall indicates unregulated aldosterone production. In the other two tests shown here, IV or oral saline load, an excess of sodium is given, which should suppress aldosterone production through its negative feedback loop. Failure of the aldosterone to fall again indicates unregulated aldosterone production.

CASE #1

Returning to the patient, an abdominal CT scan reveals an ovoid nodule in the right adrenal gland, 2.1 cm in greatest dimension, as shown below:

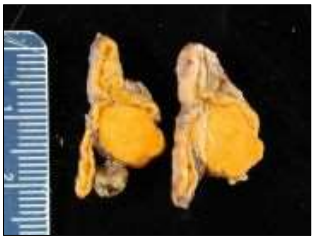


The patient is scheduled for an adrenalectomy.

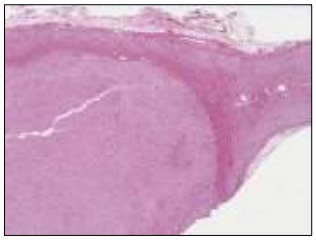
Imaging reveals what we expected: there is a 2.1-cm mass in the adrenal gland, which is presumably producing aldosterone. (The imaging here is subtle, and you wouldn't be expected to recognize this mass on your own at this point.) Let's get that tumor out with an adrenalectomy.

CASE #1

As the endocrine pathologist on service, you review the gross specimen and microscopic slide linked below:



Adrenal VS 1-1



Adrenal VS 1-2

Take five to ten minutes to examine the two virtual slides and answer the questions on the following four PowerPoint slides.

CASE #1

Examine Adrenal VS 1-1 and answer the following questions:

Q9: Before considering the mass, let's take stock of the normal structure of the adrenal gland. What are the red (A) and yellow (B) arrows pointing to? What is being indicated with the blue arrow (C)?

The red arrow points at the adrenal medulla, and the yellow arrow to the adrenal cortex. The adrenal medulla tends to have a more red-brown to tan appearance, whereas the cortex has a bright golden yellow to orange hue. The blue arrow indicates the central vein running through the gland. There are cross-sections of additional, smaller medullary veins visible in the specimen.

Q10: Based on the appearance of the mass, indicated by the pink circle, do you favor this to be a benign or malignant neoplasm? What features support this hypothesis?

This mass is likely benign, given that it is well-circumscribed and uniform in appearance, has "pushing" borders, and has no evidence of necrosis or hemorrhage. There is no evidence of invasion into the surrounding tissue or the surrounding capsule of the organ.

The adrenal gland is really *two* glands that are just anatomically associated. The **adrenal cortex** is the outer portion of the gland and is responsible for generating several **steroid hormones** (glucocorticoids, like cortisol; mineralocorticoids, like aldosterone; and sex hormones). The **adrenal medulla** is the inner portion of the gland is responsible for generating **catecholamines**, like epinephrine and norepinephrine. The medulla contains several medullary veins; the largest, running down the central axis of the gland, is the **central vein**.

We've looked at many tumors at this point through several organ systems, so hopefully you're feeling comfortable with features of benign versus malignant tumors. **Benign** tumors are noninvasive, so tend to be well-circumscribed and have "pushing" borders (rather than infiltrative/invasive borders). They usually do not have notable necrosis or apoptosis. These features are all seen in the gross specimen, suggesting that this tumor is benign.

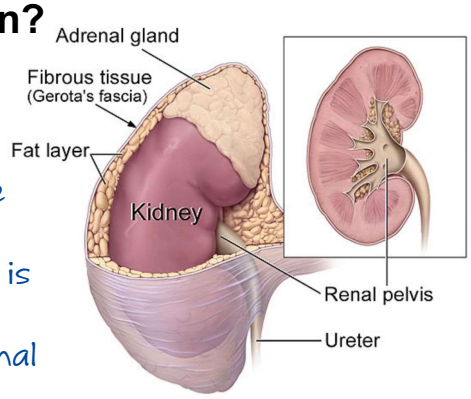
CASE #1

Continue to examine Adrenal VS 1-1 and answer the following questions:

Q11: What color is the mass, indicated again with the black asterisk? What does this tell you about the likely origin and chemical composition of the cells that make up this tumor?

The mass is bright, golden yellow in color and matches perfectly with the color of the background adrenal cortex. Accordingly, it likely arises from this tissue. The bright yellow color represents intracellular lipid. Categorically, the adrenal cortex produces steroid hormones, which use cholesterol (a lipid) as a precursor molecule.

Q12: What is indicated with the white polygon? In what way does this tissue reflect the anatomic location of the adrenal glands?



This is adipose tissue and is an example of some of the retroperitoneal fat in which the adrenal glands are embedded. More specifically, this is perirenal fat that is contained within Gerota's fascia; the perirenal fat surrounds not only the kidney but the ipsilateral adrenal gland as well.

The mass itself is bright golden-yellow, just like the cortex. This implies that it came from the adrenal cortex, too. As mentioned previously, the predominant function of the adrenal cortex is to produce steroid hormones. These are lipid-derived, coming from cholesterol, so it makes sense that the tissue stains a bright yellow color like other lipid-containing tissues in the body.

The adrenal glands sit atop the kidneys bilaterally and are embedded in perirenal fat and enclosed within **Gerota's fascia**. This perirenal fat represents just a subportion of the retroperitoneal fat.

CASE #1

Examine Adrenal VS 1-2 and answer the following questions:

Q13: At low power, does the histological appearance match that of the gross specimen? Does the architecture of the mass suggest that this is a benign or malignant process?

Yes, this has a similar appearance: the mass is easy to distinguish from the background tissue, has a relatively clear "pushing" border and no evidence of necrosis or hemorrhage. This remains likely a benign process.

Q14: Zooming in to the edge of the mass, as indicated by the blue line, is it well-circumscribed? Is it encapsulated? What's the difference?

Yes, this tumor is well-circumscribed: the edge of the lesion is well defined and easy to discern from the surrounding tissue. But it is not encapsulated; a tumor capsule contains fibrous tissue and will completely or partially surround a tumor. Here, there is no notable fibrous tissue surrounding the tumor.

Now we move to the histology of this mass. At a microscopic level, we see the same themes from before that suggest it to be a benign: easy distinction from the surrounding tissue without evidence of infiltration or invasion, "pushing" borders, and no evidence of necrosis. But note that the tumor itself, while well-circumscribed (easy to discern from the surrounding tissue), is *not* encapsulated (surrounded by fibrous tissue).

CASE #1

Continue to examine Adrenal VS 1-2 and answer the following questions:

Q15: Examine the tumor cells encircled in yellow. Does their appearance match with your prediction of the source of the tumor? Based on the clinical findings, from what specific sublayer did these cells presumably originate?

These cells are relatively pale and appear to have much of their cytoplasm washed out—this is consistent with the prediction that they would contain lipid, and that they would have arisen from the adrenal cortex. More specifically, given the patient's findings of hyperaldosteronism, they likely came from the mineralocorticoid (aldosterone)-producing cells of the zona glomerulosa.

Q16: What are the structures indicated with the red arrows? What is their role in the pathophysiology of this tumor?

These are blood vessels. In general, tumors that are metabolically active (in this case, producing hormones) require sufficient perfusion to accomplish their cellular functions. In addition, that hormone must be secreted into the blood stream to cause the systemic, end-organ effects seen in this tumor.

Alright, let's take this story home. Looking at the neoplastic cells, they have pale, vesicular, eosinophilic cytoplasm. The paleness comes from lipid in the cytoplasm having washed out during processing, and we expect this—it's a tumor producing steroid hormones. We previously mentioned the adrenal cortex as the source of these hormones, but we can subdivide it into three layers:

- The **zona glomerulosa** produces **aldosterone** and is the most superficial of the three layers. Given the chemistry of this patient's tumor, it is presumably derived from this layer.
- The **zona fasciculata** produces **cortisol** and is the next deepest.
- The **zona reticularis** produces **sex hormones** and is the deepest.

Within the tumor, we see the cross-sections of many, many small blood vessels. This is a standard finding in endocrine tissues: the way they work is by secreting their substances into the blood stream to distribute and travel to target tissues. Thus, tumors that derive from these tissues also generally have rigorous blood supply.

PRIMARY HYPERALDOSTERONISM: SUMMARY

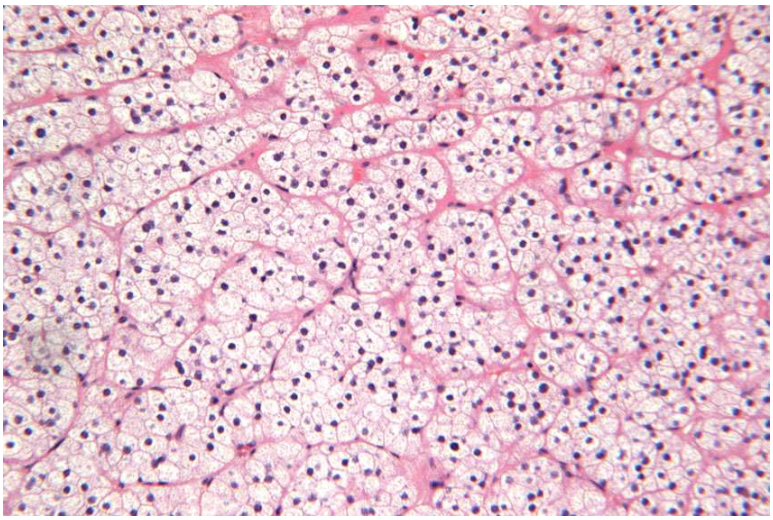
The patient is diagnosed with an adrenal cortical adenoma (aldosteronoma). Their blood pressure trends down following surgery, and they experience no recurrence. They discontinue pharmacotherapy for hypertension.

Let's review key features of the gross and histopathology of this lesion.



gross appearance

well-circumscribed, golden yellow tumor in adrenal cortex



histopathology

nests and cords of clear to pale eosinophilic, vesicular cells with fine intervening blood vessels

Finished up our first case. We've cured the patient! (Removal of a benign tumor generally will.) As a notable benefit, they also don't have to be on as many medications as previously as their blood pressure trends down into the normal range. This reduces risk of additional adverse effects from those medications.

Briefly, as a benign tumor, we expect to see a well-circumscribed, uniform tumor without necrosis or hemorrhage. The gross image also demonstrates the golden yellow to orange color of the tumor, indicating the lipid content of the tumor cells. That can also be seen microscopically through the relatively clear cells (with pale, vesicular eosinophilic cytoplasm). Also present in the tumor are many small capillaries, carrying blood to the tumor cells and picking up the aldosterone they secrete.

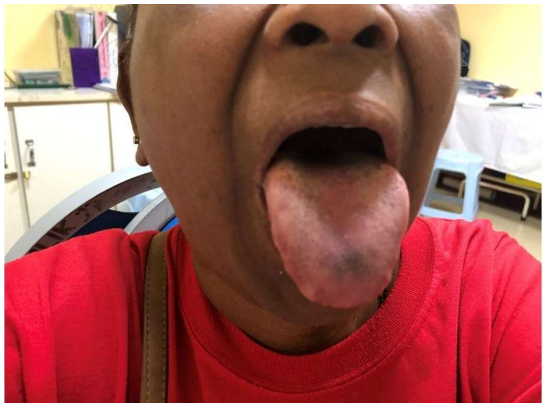
Case #2

CASE #2

A 32-year-old patient presents to the Emergency Department with severe fatigue, dizziness, and vomiting for the past three days. They report feeling increasingly weak and lightheaded. The patient also mentions darkening of their skin and unintentional weight loss over the past few months.

Past medical history is notable for type 1 diabetes, diagnosed at age 9.

On examination, the patient is profoundly fatigued and dehydrated. Vitals are as follows: T 98.9°F, BP 100/62 mmHg, HR 120, RR 18, SpO_2 99% (on room air). The oral and skin examinations reveal the following:





Our next case focuses on a patient with marked hypotension and clinical findings of fatigue, dizziness, vomiting, weakness, and lightheadedness. One unusual symptoms they complain of is a change in complexion: their skin—and some mucous membranes—have become hyperpigmented. The past medical history is notable: this patient already carries the diagnosis of an autoimmune disease (type 1 diabetes). The vital signs demonstrate hypotension and tachycardia, the latter likely being a reactive change to the low blood pressure. The patient is somewhat tachypneic, too.

CASE #2

Q1: What do you see in the physical examination?

There is evidence of hyperpigmentation, not only in areas that are normally pigmented (like the skin), but also the mucous membranes of the oral cavity (especially on the dorsal tongue, as shown in the image to the left in this case). The mucous membranes also appear dry; there is no glistening to the tongue (though this would be more evaluable with a live patient and not just a static image in front of us). The distal digits and skin overlying the knuckle joints show hyperpigmentation in the image on the right.

Appropriate labs will be drawn, but let's consider the patient's current clinical status.

Q2: What's your next step in managing this patient?

We'll have to proceed with a workup to diagnose what is going on in this patient, but their vital signs right now are concerning. They're hypotensive, tachycardic, weak, and fatigued. They have evidence of dehydration. We have to get some fluid resuscitation going to help stabilize the patient while proceeding with the workup.

As mentioned on the previous slide, there is **hyperpigmentation** seen both in the skin and the oral mucous membranes. Note that this hyperpigmentation isn't exactly diffuse; that is, not all parts of the skin and mucous membranes are impacted. The predominant changes are overlying joint surfaces and in skin folds. The hyperpigmentation in this disease is often described as "bronzing" based on the specific hue of hyperpigmentation—but that term is restrictive, as it would only apply to lighter-skinned individuals.

This patient is unwell and clearly has evidence of dehydration. The vital signs are abnormal. There aren't significant findings suggesting volume overload in this patient (ascites, peripheral edema, etc.) so we need to get some fluids into the patient urgently. We'd start an intravenous line and get boluses of fluid into the vascular system; at the same time, we can start collecting labs for evaluation.

CASE #2

The patient's CBC returns with an elevated hematocrit; WBC and platelets are within normal limits. The metabolic panel is resulted below:

Test	Value	Normal Range
Na	128 (**L**)	136–145 mEq/L
K	6.5 (**H**)	3.5–5.0 mEq/L
Cl	98	95–105 mEq/L
CO2	18 (**L**)	22–28 mEq/L
BUN	30 (**H**)	7–18 mg/dL
Cr	1.4 (**H**)	0.6–1.2 mg/dL
Glucose	60 (**L**)	70–110 mg/dL

Q3: How do you interpret the findings related to the electrolytes? How do they relate to the patient's vital signs?

There is evidence of hypoaldosteronemia; as seen in Case #1, aldosterone causes retention of Na^+ and secretion of K^+ and H^+. Here, we have just the opposite: sodium wasting and retention of potassium and protons (low CO_2, representing low $[HCO_3^-]$). Sodium wasting will cause dehydration as water follows the sodium out in the urine; this leads to hypotension, which reflexively causes tachycardia.

There are lots of abnormalities in the patient's labs; let's consider the different parts of this lab panel in pieces. First, we can group the abnormalities in sodium, potassium, and bicarbonate (dissolved CO_2) together: hyponatremia, hyperkalemia, and acidosis (low CO_2) can be linked together by a lack of **aldosterone** function. This hormone normally causes retention of sodium ions (and, by extension, water) while increasing the secretion of potassium ions and protons. Lack of aldosterone (or response to it) will therefore lead to loss of sodium with retention of potassium and protons. This **salt-wasting** and the associated movement of water into the urine by osmosis, leads to hypotension. And hypotension will lead to tachycardia, as the heart attempts to maintain perfusion in the body.

CASE #2

Q4: What is the differential of the hypoglycemia in this patient?

We'd like to find a unifying diagnosis, and a lack of aldosterone could easily be associated with a lack of glucocorticoids—they'd both be expected with adrenal failure. But this patient also carries a diagnosis of type 1 diabetes. Taking too much insulin (especially if the patient has been struggling to eat with as ill as they feel) could also cause, or contribute to, the hypoglycemia.

Q5: How do you interpret the BUN, creatinine, and high hematocrit?

These are all likely related through the patient's dehydration. Hypotension secondary to the dehydration may cause prerenal azotemia (note the BUN:Cr ratio >20) and ultimately lead to prerenal acute kidney injury (AKI). The high hematocrit is likely a concentration artifact from the plasma volume being decreased. Remember that hematocrit is a measure of the percentage of blood volume occupied by red blood cells; if the amount of plasma is down, the percentage that is red cells increases.

Q6: What is the likely diagnosis? How might we confirm it?

This is likely adrenal failure. Let's measure the relevant hormones from the hypothalamic–pituitary–adrenal (HPA) axis and take it from there.

This patient also has hypoglycemia. Our first intuition is to link this finding with the likely deficit of aldosterone: a deficit of **cortisol**, a **glucocorticoid**, would also be expected to lead to hypoglycemia. Think broadly, though. This patient already carries a diagnosis of type 1 diabetes, and hypoglycemia could easily be related to that underlying diagnosis, in whole or in part. Patients who take insulin (which is true for all patients with type 1 diabetes) need to titrate boluses of short-acting insulin against their oral intake. If oral intake is down but the patient is still taking insulin—hey, they feel sick and have dizziness and lightheadedness, which may make them less likely to have adequate oral intake—this disproportion may lead to hypoglycemia.

With hypotension, we have decreased intravascular volume, and that may start to lead to prerenal acute kidney injury. We have evidence of prerenal azotemia on the metabolic panel, with elevated BUN and creatinine and BUN:creatinine ratio >20. (Feel free to go back to the Renal Block for a review of renal pathophysiology.) The decreased intravascular volume here is due predominantly to a loss of plasma volume. Remember back to Hoste Defense/Blood: the hematocrit is the percentage of blood volume occupied by red blood cells. If plasma volume is down but red cell volume hasn't changed, then it occupies a larger percentage of the blood. This is a concentration artifact.

We're dancing around the diagnosis, but let's put the pieces together. This is an example of **adrenal failure** (**adrenal insufficiency** or **adrenal crisis** may also be used).

CASE #2

Additional lab tests are ordered the show the following:

Test	Value	Normal Range
Serum cortisol (8 am)	<1.0 mcg/dL	5–23 mcg/dL
Plasma aldosterone concentration	<1 ng/dL	7–30 ng/dL
Plasma ACTH (8 am)	>2000 pg/mL	10–60 pg/mL

Q7: What do these findings suggest about the anatomic location of this patient's problem?

This is a problem at the level of the adrenal cortex (primary adrenal insufficiency), not at the level of the anterior pituitary. The low cortisol and aldosterone concentrations indicate that there is adrenal insufficiency; the markedly elevated plasma ACTH demonstrates that the anterior pituitary is *trying* to stimulate the adrenal cortex but is unsuccessful at doing so.

Q8: If the labs had been suggestive but less definitive, then what?

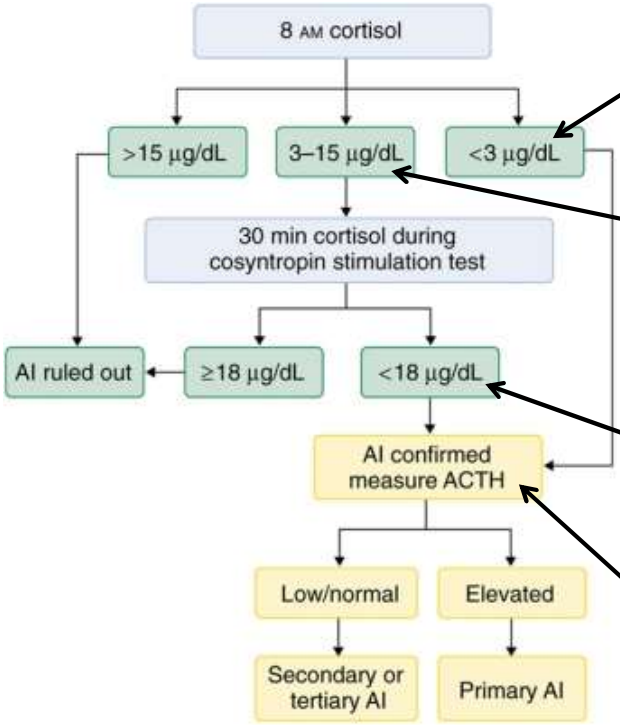
A cosyntropin stimulation test may rule in/out the diagnosis of adrenal insufficiency.

To investigate the function of the adrenal, we'll measure relevant hormones in the **hypothalamic–pituitary–adrenal** (**HPA**) **axis**. While **corticotropin-releasing factor** (the hormone secreted by the hypothalamus in this pathway) is not routinely measured, the hormones from the pituitary and adrenal are. Normally, **adrenocorticotropic hormone** (**ACTH**) should be released by the anterior pituitary and simulate the adrenal cortex to produce **cortisol** and **aldosterone**. Here, we see what happens if the adrenal fails to respond. The anterior pituitary is "screaming" at the adrenal gland with a markedly elevated ACTH level, but the gland isn't responding. We see cortisol and aldosterone concentrations below the measurable range. This implies that the problem is at the adrenal gland (primary adrenal insufficiency); it's not a failure of ACTH production and secretion (secondary adrenal insufficiency).

These lab results are diagnostic of **adrenal insufficiency**, and no further workup would be needed to diagnose this condition medically. But let's consider what to do if the case isn't quite so "cut and dried."

CASE #2

Q9: How does a cosyntropin stimulation test work?



our patient's cortisol was low enough that the diagnosis of adrenal insufficiency (AI) was never in doubt

but a milder case may not have low cortisol levels yet, so let's see how well the adrenal cortex responds to synthetic ACTH (cosyntropin)

if the adrenal's response is weak, that also confirms adrenal insufficiency

and we can distinguish if the problem is in the anterior pituitary (or hypothalamus) vs. the adrenal cortex using endogenous ACTH level

We can see the scenario our patient was in: 8 AM cortisol was below the measurable range, certainly below 3 mcg/dL. (Cortisol is measured at 8 AM because this hormone has a circadian cycle and is highest early in the morning.) In conjunction with the elevated ACTH, that's diagnostic of **primary adrenal insufficiency**.

But if the 8 AM cortisol had been low but not quite *that* low, we can consider a **cosyntropin stimulation test**. In this test, a bolus of cosyntropin, a synthetic version of ACTH, is given and the concentration of cortisol is measured again (usually 30 or 60 minutes later). If the gland fails to respond to the synthetic ACTH, we can rule in adrenal insufficiency. If it does respond with an increase in cortisol concentration, we can rule out adrenal insufficiency. You don't need to know the exact cutoffs at this point but should appreciate the general concepts behind how this test works.

CASE #2

Let's return to the patient and deal with their acute problems.

Q10: So what is the diagnosis? What additional therapy should be initiated immediately?

> This is acute adrenal insufficiency, or adrenal crisis. The patient must be started on intravenous or intramuscular steroids; hydrocortisone is standard of care.

The patient is managed appropriately; their vital signs, electrolytes, and other laboratory parameters return to normal.

Q11: What are some causes of primary adrenal insufficiency (Addison disease)? Which is most likely in this patient?

> Most common in the U.S. and for this patient is autoimmune destruction. Worldwide, the most common cause is tuberculosis. Other causes include abrupt cessation of exogenous steroids, bilateral adrenal hemorrhage (sometimes seen in certain causes of sepsis), tumors (primary or metastatic), amyloidosis, genetic disorders (adrenoleukodystrophy, congenital adrenal hyperplasia).

We gave the diagnosis previously, but just to declare it again: this is a case of **acute adrenal insufficiency** (**adrenal crisis**). We've already started the patient on fluids; now we should also replete the steroid hormones that are lacking. The usual drug of choice is **hydrocortisone**, which has both glucocorticoid and mineralocorticoid effects. It can be administered intravenously or intramuscularly.

So, what caused this patient's adrenal failure (**Addison disease**)? Statistically, in the U.S., the most common cause of Addison disease is autoimmune destruction of the adrenal glands. Worldwide, the most common cause is tuberculous infection of the glands. A major cause to consider is abrupt cessation of steroids used in the treatment of other autoimmune diseases, inflammatory disorders, and several other diagnosis. After chronic use of steroids, they must be carefully tapered to avoid adrenal crisis. This is because the adrenal gland stops producing its own hormones when an exogenous source is provided, and it takes time to ramp back up endogenous hormone production. A few other examples are included in the subsequent list, one additional one of which we'll explore in the virtual slides.

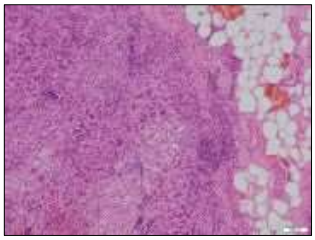
It's also worth mentioning that **secondary adrenal insufficiency** can also occur when there is a decrease in ACTH release from the anterior pituitary. This is usually due to **hypopituitarism**, where the gland may be damaged by a tumor within it, infarcted, or otherwise injured.

CASE #2

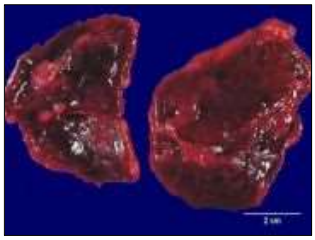
This patient would *not* undergo a biopsy or resection, but as the endocrine pathologist on service, you consider what the patient's adrenal glands might look like.



Adrenal VS 2-1



Adrenal VS 2-2



Adrenal VS 2-3

Take five to ten minutes to examine the three virtual slides and answer the questions on the following three PowerPoint slides.

CASE #2

Examine Adrenal VS 2-1 and answer the following questions:

Q12: Given the likely etiology of the patient's adrenal insufficiency, could their adrenal glands look like the ones in the black (solid) box? If not, could ANY patient whose glands look like this have adrenal insufficiency?

This is likely the appearance of our patient's adrenal glands: they are atrophic (shrunken compared to normal). This is due to autoimmune destruction of the gland.

Q13: Given the likely etiology of the patient's adrenal insufficiency, could their adrenal glands look like the ones in the yellow (dashed) box? If not, could ANY patient whose glands look like this have adrenal insufficiency?

These adrenal glands are not consistent with our patient's likely autoimmune etiology but could be seen with adrenal insufficiency due to congenital adrenal hyperplasia (especially 21-hydroxylase deficiency). The glands are hyperplastic but fail to produce cortisol and aldosterone.

Of note, the appearance of the adrenal glands could be highly variable and still result in the same finding of adrenal insufficiency. Our patient's glands are most likely atrophic, having been damaged and destroyed by autoimmune attack; this is the most common cause of Addison disease in the United States. That gland destruction will lead to thinning and eventual loss of the adrenal cortex, as seen in the top glands in the virtual slide.

But the adrenal glands could look just the opposite and still lead to the same clinical findings. An example for this would be **congenital adrenal hyperplasia** (**CAH**), a pediatric disease that is due to a failure of one of the enzymes in steroid hormone production. With 21-hydroxylase deficiency, there is an inability to metabolize steroid precursors to cortisol and aldosterone. This leads to an increase in ACTH production through the negative feedback system in the HPA axis (no cortisol or aldosterone will lead to upregulation of ACTH production in an attempt to increase cortisol and aldosterone concentrations). That increased ACTH leads to hyperplasia of the glandular cells, who are receiving increased trophic signaling. They just can't produce the mature hormones.

CASE #2

Examine Adrenal VS 2-2 and answer the following questions:

Q14: What is the predominant inflammatory cell involved in this disease, as shown in the yellow box?

There are predominantly lymphocytes, forming aggregates of small round nuclei.

Q15: Some cells of the adrenal cortex that still viable actually undergo hypertrophy and/or hyperplasia, as shown in the blue circle. What drives this?

While many of the cells of the adrenal cortex are dying off, the levels of cortisol drop. Through reduction of feedback inhibition to the hypothalamus and anterior pituitary, there is an increase in ACTH concentration. The viable cells react to this increased ACTH concentration by increasing the amount of hormone they produce. This ACTH stimulation causes them to become hypertrophic and/or hyperplastic, but eventually they cannot keep up with demand and adrenal insufficiency ensues.

At a microscopic level, we see what we might expect in an autoimmune disease. There is infiltration of the adrenal cortex with lymphocytes, both individually and in small aggregates. In similar fashion to congenital adrenal hyperplasia, described on the previous slide, the cells of the cortex that are still viable may become hypertrophic and/or hyperplastic because they are receiving excessive ACTH signaling. But they cannot make up for all the other cells that have been lost, and adrenal insufficiency still occurs.

CASE #2

Examine Adrenal VS 2-3 and answer the following questions:

Q16: These adrenal glands came from the autopsy of another patient who suffered acute adrenal crisis. What do you observe at the tips of the red arrows?

There is hemorrhage in both adrenal glands—enough to stain both the parenchyma and the perirenal fat in which the glands are embedded.

Q17: What is this syndrome? What are some common causes?

This is Waterhouse–Friderichsen syndrome: bilateral adrenal hemorrhage. It is a potential complication of severe sepsis with several different organisms. A few that are commonly associated include:

- *Neisseria meningitidis* (most common)
- *Pseudomonas aeruginosa* (often in children)
- *Streptococcus pneumoniae*
- …and many others

Noninfectious causes of bilateral adrenal hemorrhage include anticoagulant medications and disseminated intravascular coagulation (DIC).

At first glance, the predominant change seen in these adrenal glands is a color change: they're entirely red (as is the surrounding fat). This is overwhelming hemorrhage that destroys the two glands, called **Waterhouse–Friderichsen syndrome**. The most common association is sepsis related to bacterial meningitis caused by *Neisseria meningitidis*, although many other bacteria and viruses have also been implicated. Two others are listed here; a case series in children dying of bilateral adrenal hemorrhage showed *Pseudomonas aeruginosa* to be the most common pathogen in that population. *Streptococcus pneumoniae* and many other bacteria classically associated with sepsis may also lead to this complication.

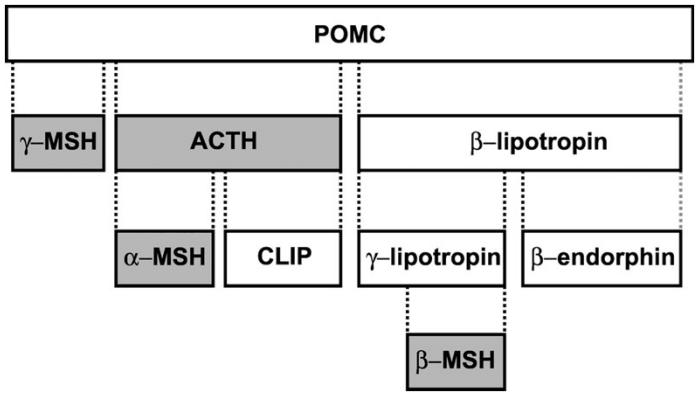
There are also noninfectious reasons—things *besides* Waterhouse–Friderichsen syndrome—why one might develop bilateral adrenal hemorrhage. These include the use of anticoagulants, wherein this would be an adverse event. It also includes disseminated intravascular coagulation (DIC).

CASE #2

The patient has some follow-up questions:

Q18: "What was going on with my skin and tongue? Why did they get darker?"

Several of the melanocortins (melanocyte-stimulating hormones, MSH) come from the same precursor as ACTH, called proopiomelanocortin (POMC). Upregulation of ACTH production may increase the levels of these other melanocortins. One of these melanocortins, α-melanocyte stimulating hormone, comes from ACTH. In high concentration, ACTH can have cross-reactivity and stimulate melanocytes, increasing pigmentation.



Returning to the somewhat unusual symptom of hyperpigmentation, let's get a pathophysiological understanding of where this comes from. We know this patient's production of ACTH is upregulated, due to a decrease in the negative feedback to the hypothalamus and anterior pituitary. ACTH itself comes from a precursor molecule called **proopiomelanocortin** (**POMC**). This precursor is further metabolized into multiple hormones, including some **melanocortins** (hormones that act on melanocytes). In fact, ACTH itself can be metabolized into a melanocortin called ***α*-melanocyte stimulating hormone** (***α*-MSH**). And ACTH itself has enough chemical similarity with this metabolite that it, itself, can also stimulate melanocytes. The net result? An upregulation in ACTH production concomitantly results in an increase in melanocortins, and increased skin (and potentially mucous membrane) pigmentation results.

CASE #2

Q19: "Could this be related to my type 1 diabetes?"

It certainly could be. There are several autoimmune polyendocrine (polyglandular) syndromes (APS); the most common is APS-2, which would potentially match our patient. Further workup and screening in other organ systems may be warranted.

Table 1. Classification and Characteristics of Autoimmune Polyendocrine Syndromes.*

Characteristic	APS-1	APS-2	IPEX
Main manifestations	Addison's disease, hypoparathyroidism, chronic mucocutaneous candidiasis	Addison's disease, autoimmune thyroid disease, type 1 diabetes	Autoimmune enteropathy, neonatal type 1 diabetes, eczema
Other, associated manifestations	Primary ovarian insufficiency, autoimmune thyroid disease, type 1 diabetes, gastritis, enteritis with malabsorption, hepatitis, pancreatitis, pneumonitis, nephritis, vitiligo, alopecia, nail dystrophy, enamel hypoplasia, keratitis, retinitis	Autoimmune gastritis, alopecia, vitiligo, celiac disease, primary ovarian insufficiency	Autoimmune thyroid disease, hemolytic anemia, thrombocytopenia
Typical age at onset	Childhood, adolescence	Adolescence to adulthood	Infancy
Prevalence	1:100,000	1:1000	1:1,000,000
Treatment	Hormone replacement, antifungal therapy, immunosuppressive therapy for hepatitis, malabsorption, nephritis, pneumonitis, keratitis	Hormone replacement	Hormone replacement, bone marrow transplantation
Complications, including death	Adrenal and hypocalcemic crises, cancer in mouth and esophagus	Adrenal crisis, complications of diabetes	Infections
Genes and mode of inheritance	*AIRE*, autosomal recessive and dominant	Polygenic: MHC and others	*FOXP3*, X-linked
Immune phenotype	Autoantibodies against interferon-ω and interferon-α (>95%), organ-specific intracellular proteins	Autoantibodies against 21-hydroxylase, GAD65, IA-2, thyrotropin receptor, TPO	Autoantibodies against GAD65, lymphocytosis, eosinophilia, overproduction of cytokines, hyper IgE

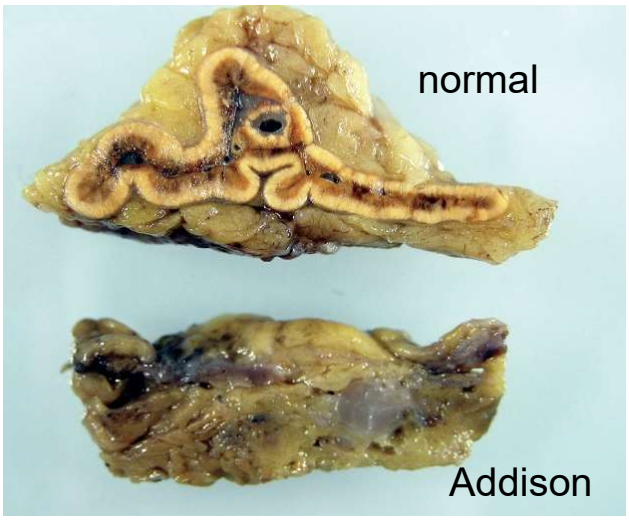
* AIRE denotes autoimmune regulator; APS-1 autoimmune polyendocrine syndrome (APS) type 1; APS-2 APS type 2; FOXP3 forkhead box P3; GAD65 glutamic acid decarboxylase 65; IA-2 islet antigen 2; IPEX X-linked immunodysregulation, polyendocrinopathy, and enteropathy; MHC major histocompatibility complex; and TPO thyroid peroxidase.

This patient's history was notable for type 1 diabetes, an autoimmune disease that results in destruction of the insulin-producing β-cells in the pancreatic islets of Langerhans. In general, having one autoimmune disease may increase one's risk of developing another autoimmune disease, likely due to underlying immunological dysregulation. There is often overlap of autoimmune diseases in the endocrine system, including a few **autoimmune polyendocrine** (**polyglandular**) **syndromes**. The specific associations are listed here; you are NOT expected to memorize this chart by any stretch (at least not for the Pathology Thread) but should appreciate the co-occurrence of multiple autoimmune diseases in these syndromes. Our patient might have autoimmune polyendocrine syndrome, type 2 (APS-2) given the history of both Addison's disease and type 1 diabetes; they also are in the right age group for this diagnosis (in fact, they technically fulfill the clinical criteria for diagnosis, having two of the three main manifestations of the disorder). Recognizing these potential syndromic associations is important for ongoing screening in other organs that may be affected. (We may want to run a thyroid panel on this patient, for example.)

PRIMARY ADRENAL INSUFFICIENCY: SUMMARY

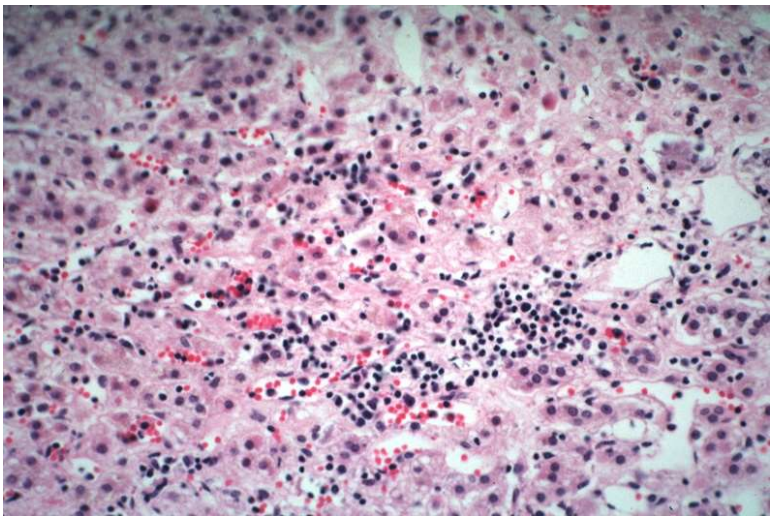
The patient's is ultimately discharged on oral glucocorticoid and mineralocorticoid replacement and does well.

Let's review key features of the histopathology of this lesion, when due to autoimmune etiology (most common in the U.S.).



gross appearance

(bilaterally) atrophic adrenal glands, particularly affecting the adrenal cortex



histopathology

lymphocytic infiltration; varying degrees of atrophic and hypertrophic/hyperplastic change

Alright, we've completed another case. This patient cannot be cured but can be treated with hormone replacement for the rest of their life.

Most of this case focused on the physiology and pathophysiology of the hypothalamic–pituitary–adrenal (HPA) axis and less on the gross and microscopic pathology for two reasons. One, if we understand the mechanism of the disease (autoimmune gland destruction), the gross findings (atrophy) and histopathology (chronic inflammation) flow very naturally from that understanding. We add the slight detail that living cells in the adrenal may show hypertrophic or hyperplastic change in response to the upregulation of ACTH. Second, this isn't a disease that we diagnose with a biopsy or resection, unless the patient died acutely, and the diagnosis is made postmortem in an autopsy.

Case #3

CASE #3

A 36-year-old patient presents to their primary care physician with left upper limb numbness and hypertension. The patient was in their usual state of health until one month prior, when the patient was in a minor car accident and had an MRI of their cervical spine performed because there was concern for whiplash. This MRI showed a cystic tumor with a mural nodule in the cervical spinal cord. Their blood pressure was noted to be elevated to 152/100 and they were told to follow-up with their primary care physician for further management and potential referral to a neurosurgeon.

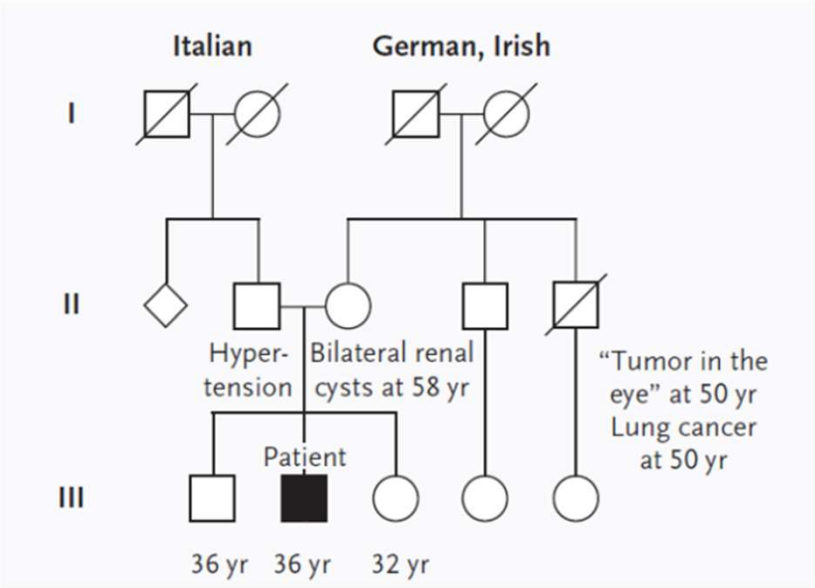
At this appointment, the patient complains of severe night sweats that often necessitate changing their sheets. They feel very anxious and feel like their heart is racing and "going to explode" several times a day. "Are these panic attacks?" they ask. "They come out of nowhere, last for a few minutes where I feel like I'm going to die, and then they're just, like, gone."

The patient's past medical history is unremarkable. They take no medications and deny use of any drugs of abuse.

Our third patient, like our first, comes in with hypertension. But there are associated episodes of severe anxiety, tachycardia, and "feeling of doom." The patient also has some neurological symptoms that are likely related to the incidental finding of a cystic tumor in the cervical spinal cord. We'll have to consider how these two entities, seemingly unrelated, may connect to each other. It's important to rule out potential exposures.

CASE #3

The patient's family history includes the diagnoses in the pedigree below:



Q1: What does the family history suggest about the diagnosis?

While it does not *have* to be related, we should consider that the patient's problems have an inherited, genetic cause.

Well, there's quite the family history in this patient. The patient's father also had hypertension, and their mother had bilateral renal cysts. One of the mother's siblings, the patient's uncle, had an eye tumor (otherwise undescribed) and lung cancer at a relatively young age. He is deceased (hence the slash in the pedigree). Multiple cancers and kidney disease in the family history, and now with a tumor in an unusual location? We should be thinking there's something heritable going on here—potentially a cancer-related syndrome?

CASE #3

The patient's physical exam is as follows:

- VS: T 98.8°F, P 82, RR 14, BP 140/100
- HEENT: No proptosis; thyroid normal size and consistency
- Chest: No abnormalities
- Abdomen: Soft, nontender, without organomegaly or mass
- Neurologic exam: 5/5 hand and leg strength bilaterally. Decreased sensation to pain and temperature the left upper limb.

The PCP starts the patient on an α-blocker for hypertension while awaiting the results of labs and imaging.

Q2: What lab tests and imaging might be most informative here?

Labs: urine and serum catecholamines, urine and serum metanephrines

Imaging: Abdominal CT or MRI (CT preferred for this particular diagnosis), MIBG scintigraphy

The exam demonstrates hypertension. The thyroid is normal, making hyperthyroidism less likely (although hyperthyroidism can occur without notable physical exam findings). The neurological exam demonstrates some specific sensory loss in the upper limb, suggesting involvement of the anterolateral system (spinothalamic pathway) discussed in the Neuroscience/Psychiatry Block. Remember, loss of only certain types of sensation without head and neck involvement suggests the problem is in the spinal cord. Plus, we already know the patient has a tumor located there based on the MRI findings.

The patient is temporized with antihypertensives; we'll see why the physician prescribed an α-blocker specifically a bit later.

The episodic nature of the patient's symptoms suggest a particular diagnosis. We'll name it later, but for now, the appropriate test would be the measurement of **catecholamines** (both urine and serum) and **metanephrines** (again, both urine and serum). Given the likelihood of the patient's diagnosis being related to the adrenal medulla, some abdominal imaging would be a good choice. We'll also consider the scintigraphy test on a subsequent slide.

CASE #3

The labs return as follows:

Test	Value	Normal Range
Urine dopamine	219 mcg/24 hr	65–400 mcg/24 hr
Urine epinephrine	9.3 mcg/24hr	0–20 mcg/24 hr
Urine metanephrine	1530 mcg/24 hr (**H**)	200–614 mcg/24 hr
Urine norepinephrine	235 mcg/24 hr (**H**)	15–80 mcg/24 hr
Urine normetanephrine	1411 mcg/24 hr (**H**)	111–419 mcg/24 hr
Urine vanillylmandelic acid	6.0 mg/24 hr	<8 mg/24 hr
Serum metanephrine	<0.20 nmol/L	<0.50 nmol/L
Serum normetanephrine	4.49 nmol/L (**H**)	<0.90 nmol/L

Q3: Do these findings explain the patient's symptoms?

There are elevated levels of catecholamines, which will have sympathomimetic effects by activating adrenergic receptors. Certainly, these levels help explain the episodic palpitations, "feeling of doom," and what is otherwise questioned by the patient to be "panic attacks." But they don't explain the neurological findings or the findings on the cervical spine MRI.

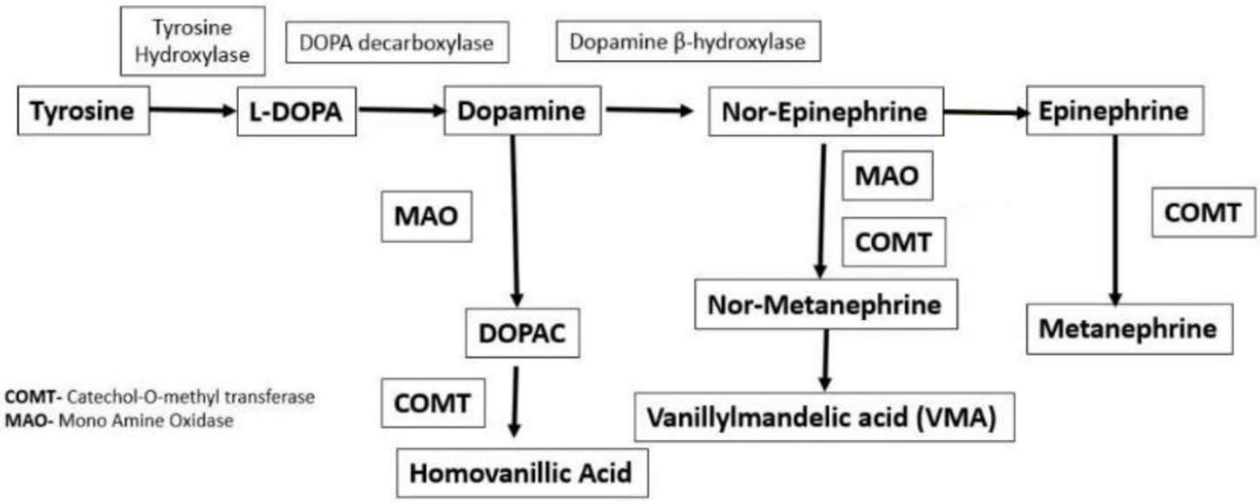
Here are our urine catecholamines and urine and serum metanephrines. Catecholamines—specifically **epinephrine** and **norepinephrine**—activate the **adrenergic receptors** of the **sympathetic nervous system**. The severe sympathetic activation these patients experience manifest as an intense fight-or-flight response: the heart rate is markedly elevated (and may be associated with palpitations), the patient is often tachypneic, feels a "sense of impending doom" (marked dread that they are about to die), and is overwhelmingly anxious.

CASE #3

Q4: What are catecholamines? What are metanephrines?

The catecholamines are a group of neurotransmitters used in the nervous system that are also produced by the adrenal medulla. They include dopamine, norepinephrine (noradrenaline), and epinephrine (adrenaline).

The metanephrines are metabolites of some of these catecholamines. They include normetanephrine and metanephrine.

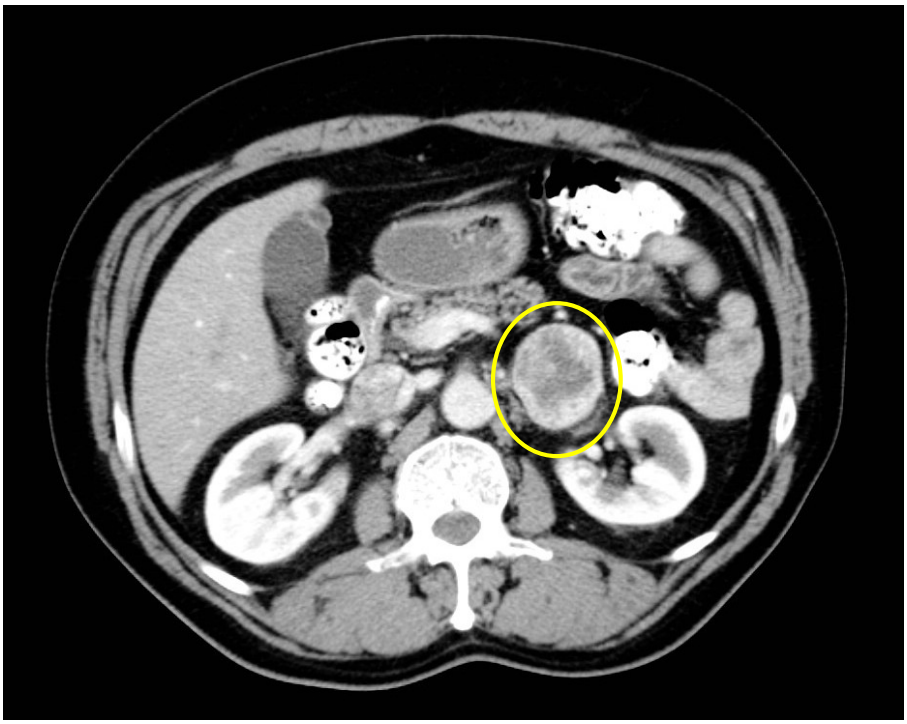


Let's consider the chemistry of these catecholamines. Categorically, **catecholamines** are **neurotransmitters** and include **dopamine**, **epinephrine**, and **norepinephrine**. In the sympathetic nervous system, norepinephrine is used by the postganglionic neuron to stimulate its effector, which has **adrenergic receptors** (α_1, α_2, β_1, and β_2). Remember that the postganglionic neuron cell bodies in the sympathetic nervous system are in the **paravertebral ganglia** forming the **sympathetic trunk** (or, in some cases, in prevertebral ganglia in the abdominal cavity). The **adrenal medulla** is functionally a displaced paravertebral ganglion, where its parenchymal cells just release epinephrine and norepinephrine directly into the bloodstream (as opposed to releasing it from an axon directly onto an effector).

Catecholamines are metabolized in the circulation by the enzymes **monoamine oxidase** (**MAO**) and **catechol-O-methyl transferase** (**COMT**). Relevant to this case, norepinephrine is metabolized to **normetanephrine**, and epinephrine is metabolized to **metanephrine**. Those metabolites can be measured in the circulation; elevations in their levels imply a previous elevation of norepinephrine and/or epinephrine.

CASE #3

The patient undergoes a CT scan and a [^{123}I]metaiodobenzylguanidine (MIBG) scan. [^{123}I]MIBG is a radioisotope avidly taken up by catecholamine-secreting tissues and be evaluated in this nuclear medicine scintigraphy test.





For the imaging studies, we see a CT scan that demonstrates a mass in the left adrenal gland. The nuclear medicine scan also highlights the left adrenal gland in black. This test, a **[^{123}I]metaiodobenzylguanidine** (**MIBG**) **scan** takes advantage of the fact that this radiolabeled molecule will be picked up and used by tissues to synthesize catecholamines. Those areas with the radiolabeled MIBG will appear dark; we see the left adrenal well labeled, and we also see the bladder. The bladder is not catecholamine-producing, itself. That's just demonstrating the pharmacokinetic elimination of the radiolabel.

CASE #3

Q5: Given the history, lab findings, and imaging, what is the most likely diagnosis? Where?

This patient most likely has a pheochromocytoma (catecholamine-secreting tumor) arising from the left adrenal gland.

The patient is scheduled for surgery to remove the lesion.

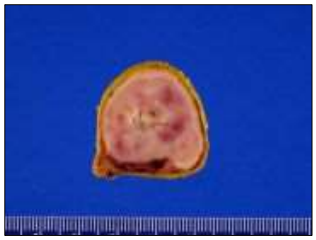
Q6: What medication(s) are appropriate—and necessary!—for the anesthesiologist to administer prior to the surgery? Does the order matter?

This patient must be given α-blockers prior to surgery. This patient has elevated circulating catecholamines, which may spike even higher as the tumor is manipulated during surgery. Catecholamines stimulate α receptors in the vasculature and cause vasoconstriction. If tachycardia occurs after α-blockade, β-blockers may also be given. However, the order should not be reversed. If β-blockers are given first, there will be unopposed activity of the α receptors, which can precipitate a hypertensive crisis.

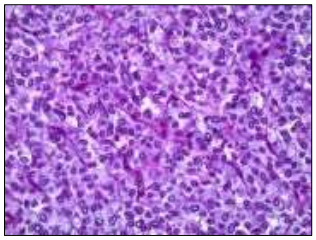
Panic attack-like episodes without clear triggers + elevated catecholamines and metanephrines + imaging and nuclear medicine studies showing a catecholamine-producing tumor in the adrenal gland? This is a **pheochromocytoma**. It must be removed surgically. But because the tumor produces catecholamines, we've got to be careful with our plan removing it—it's basically a loaded, uncapped epinephrine auto-injector (EPIPEN®) waiting to go off. The patient must be put on adrenergic receptor blockers. Most important is to block the vascular *α* receptors, because the massive amount of catecholamine that would be immediate released into the bloodstream could set off a **hypertensive crisis**. *β*-blockers may be considered if the patient's heart rate is significantly elevated, but this should only be done after *α*-blockade has been accomplished.

CASE #3

As the endocrine pathologist on service, you review the gross specimen and microscopic slide linked below:



Adrenal VS 3-1



Adrenal VS 3-2

Take five to ten minutes to examine the two virtual slides and answer the questions on the following three PowerPoint slides.

CASE #3

Examine Adrenal VS 3-1 and answer the following questions:

Q7: What portion of the adrenal gland is indicated with the yellow arrow?

This is the adrenal cortex, identifiable by its bright golden-yellow to orange color as well as its position toward the external surface of the specimen. The adrenal cortex appears thinned in the areas where it immediately surrounds the tumor. Note that this image also shows the yellow ink on the outside from the specimen being processed in the gross pathology lab. This would be for assessing margins of the tumor if it turns out to be malignant.

Q8: There is a depressed central part of the tumor, as shown with the black asterisk. This would likely have a friable (easily crumbled) texture. What is likely going on here?

There is likely central necrosis in the tumor; it has grown too fast and outpaced its blood supply, leading to ischemia and necrosis in the deepest parts of the tumor.

This gross image shows us a tan-brown, well-circumscribed but unencapsulated tumor with some areas of necrosis and hemorrhage. We see the thinned adrenal cortex on its surface, as well as the inked edge of the specimen for margin analysis.

CASE #3

Continue to examine Adrenal VS 3-1 and answer the following questions:

Q9: What does the dark maroon-red material, indicated by the red circle, likely represent?

This is likely hemorrhage. There may also be some hemorrhage within the substance of the specimen, giving some regions (say, 1:30 and 4 o'clock in the image) a bit of a darker red hue.

Q10: Given the appearance of the resection specimen, as shown in the white rectangle, what portion of the adrenal gland is likely the source of this tumor?

This has already been discussed quite a bit in this case so far, but the tumor here appears to originate from the adrenal medulla. Little, if any, normal adrenal medullary tissue is visible in this image. It has likely been compressed and damaged by the expanding tumor.

Again, we see some necrosis and hemorrhage in this tumor within the adrenal medulla. Normally, these are signs concerning for malignancy and not benignity, but that's not always the case. Here, the well-circumscribed nature of the tumor and "pushing borders" argue that it is benign. We'll also see that statistics are on the side of benign here later: about 90% of pheochromocytomas are benign.

CASE #3

Examine Adrenal VS 3-2 and answer the following questions:

Q11: How would you describe the overall architecture of this tumor?

The tumor is composed of nests of cells with fine fibrous septa containing blood vessels between the nests. This appearance is classically called "zellballen" (Ger: "cell balls") and is characteristic of pheochromocytomas and other paragangliomas.

Q12: How would you describe the appearance of the cytoplasm in the tumor cells, two of which are indicated by the red arrows? What does this appearance likely represent?

The cytoplasm is granular and basophilic. The granules themselves probably contain catecholamines, which are acidic at physiological pH and therefore stain basophilic. Note that the appearance of the cells here is not strikingly different from the normal adrenal medullary cells.

At a microscopic level, a pheochromocytoma recapitulates the cellular appearance of the adrenal medulla, but the architecture is a bit different. Cytologically, the tumor cells have granular, basophilic cytoplasm. The granules are staining basophilic because of the abundance of catecholamines within them. The nucleus, not described in the questions, shows "salt-and-pepper" chromatin (speckled), a common feature of neuroendocrine tumors throughout the body. In terms of organization, the tumor cells are arranged in nests often referred to by the German term **zellballen** ("cell balls"). In between the nests are fine fibrous septa containing delicate vessels.

CASE #3

Let's consider a bit more of the biology of this tumor.

Q13: What is a paraganglioma? Is a pheochromocytoma one?

A paraganglioma is a tumor of the neuroendocrine cells of autonomic ganglia. Sympathetic paragangliomas contain catecholamine-producing cells, which may be located inside or outside of the adrenal gland. When it is located within the adrenal gland, it may also be called a pheochromocytoma. But extra-adrenal paragangliomas would not be considered pheochromocytomas. Parasympathetic paragangliomas are usually nonfunctional (don't produce anything) and tend to exist in the head and neck.

Q14: What is the "10% rule" with pheochromocytomas?

The "10% rule" is an outdated (and oversimplified) rule of thumb that states that:

- 10% of paragangliomas are extra-adrenal
- 10% of sporadic pheochromocytomas are bilateral
- 10% of pheochromocytomas are malignant (defined entirely by the presence of metastases—the only criteria in this tumor for malignancy)
- 10% of pheochromocytomas are asymptomatic
- 10% of pheochromocytomas are hereditary (we now know this number to be closer to 25–40%)

We've called this tumor a pheochromocytoma, which categorically is a type of **paraganglioma**. Again, the adrenal medulla is essentially a displaced **paravertebral ganglion**. The generic term "paraganglioma" means a tumor of a peripheral (autonomic) ganglion. This includes sympathetic paragangliomas (which tend to produce catecholamines, resulting in the same symptoms as our patient) and parasympathetic paragangliomas (that tend not to secrete anything, so any symptoms are usually due to mass effect in the head and neck).

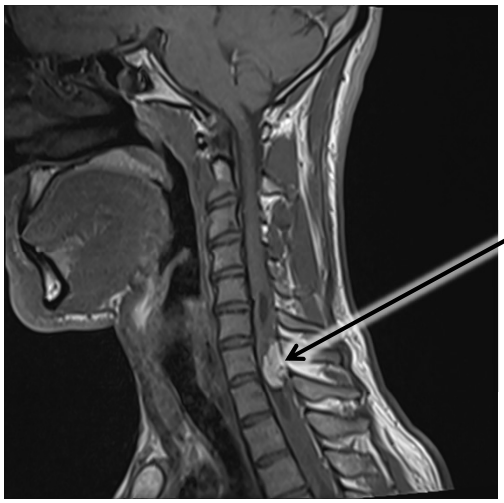
A cute mnemonic for some specific biostatical features of the pheochromocytoma is given by the "**10% rule**." In reality, there are ranges reported in the literature for each of this criteria, but it's still a useful rule of thumb. It states that about 10% of pheochromocytomas are each of: extra-adrenal, bilateral, malignant, asymptomatic, and hereditary. Extra-adrenal paragangliomas often occur in the head and neck, though sometimes intra-abdominally. Malignancy in this tumor is defined by the presence or absence of metastasis, not on the cytology or architecture of the tumor. And while it has historically been included as part of the 10% rule, the hereditary cases are much more numerous than that, on the order of 25–40% of cases.

CASE #3

The patient's hypertension and episodic symptoms improve, but the upper limb numbness and decreased pain and temperature sensation persist.

Q15: What is likely causing that symptomatology?

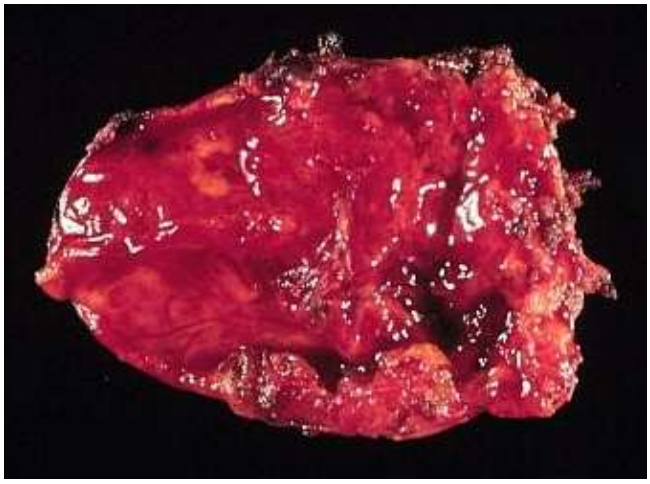
There was a mass in the cervical spinal cord. The upper limb is innervated by the spinal nerves from C5 through T1 in what is called the brachial plexus. The spinal cord mass is likely impinging on fibers running through the white matter that arise from those levels.



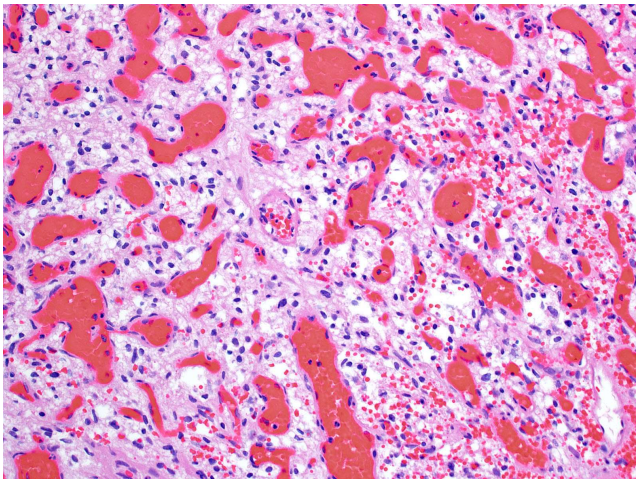
With resection of the tumor, the hypertension and panic attack-like episodes subside. But now we're left with that cystic cervical spinal cord mass and the neurological symptoms it's causing. The upper limb is innervated by the C5–T1 spinal cord levels; an issue in the spinal cord may lead to interruption of the pain and temperature pathway within the spinal cord, leading to our patient's symptoms.

CASE #3

The spinal tumor is resected by a neurosurgeon, and shows the gross and microscopic appearance below:



gross pathology



histopathology

Q16: What is the predominant tissue type you see in this specimen?

The cut surface is quite bloody, and the microscopic appearance demonstrates tons of blood vessels (with some intervening stroma, too). This is a hemangioblastoma.

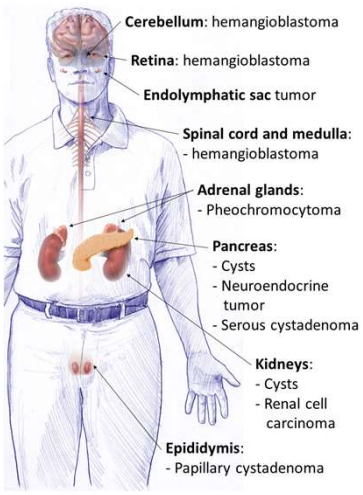
It's not our focus for today's TBL session, but the resected spinal cord tumor turns out to be a **hemangioblastoma**, a tumor of blood (*hem*–) vessel (–*angio*–) producing cells (–*blasts*). Oh, what was that cancer syndrome we learned about in the Renal Block that led to a lot of tumors with prominent vascular features?

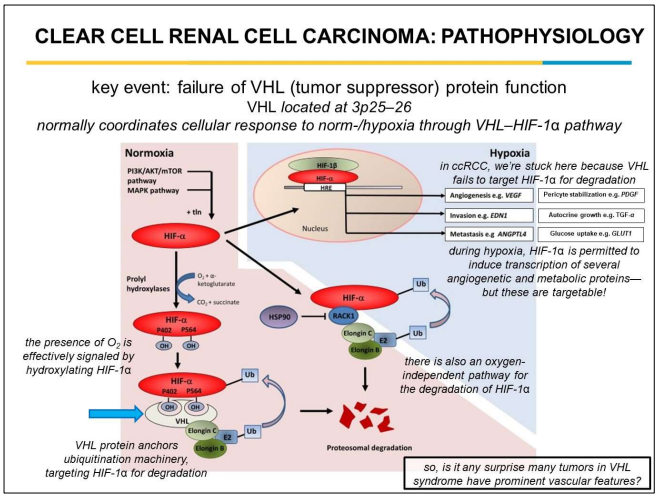
CASE #3

Combined, this family has a history of: pheochromocytoma, spinal hemangioblastoma, renal cysts, and an eye tumor (likely a retinal hemangioblastoma).

Q17: What unifying diagnosis might explain these findings?

These are classic findings of von Hippel–Lindau syndrome, an autosomal dominant disorder due to a mutation in the *VHL* gene.





Remember me from Renal?

Of course! It was **von Hippel–Lindau** (**VHL**) **syndrome**. We first learned about it in the context of clear cell renal cell carcinoma, where we mentioned that that cancer was the leading cause of mortality in these patients. Now we're seeing several of the other associated tumors. Others are listed in the diagram shown on the slide. Note again that many of the tumors have prominent vascular features, and that's because of the underlying pathophysiology. The VHL protein's primary function is to inhibit **hypoxia-inducible factor 1-*α*** (**HIF-1*α***). HIF-1*α*'s normal function is to travel to the nucleus when the cell is hypoxic and turn on several genes that would make sense for a cell experiencing hypoxia: those involved in angiogenesis (making more blood vessels) and glucose uptake (because the yield of ATP per glucose is lower under hypoxic conditions). VHL's job is to inactivate this protein when if cell is under *normoxic* conditions, when those angiogenetic and metabolic factors are unnecessary.

In VHL syndrome, one mutated copy of the gene is inherited. In the tissues, a **second hit** results in complete absence of functional VHL protein in a given cell, and that results in a failure of the breakdown of HIF-1*α*. Accordingly, the cell continuously acts as though it is under hypoxic conditions, even if it is not. This drives angiogenesis to the tumor, its uptake of glucose, as well as several other factors that allow invasion, metastasis, and autocrine growth (self-stimulating growth).

(Note: if you go back to the original pedigree for this case, you'll see lung cancer being listed in one of the relatives. This is not classically associated with VHL syndrome and may have been due to other, nongenetic causes.)

CASE #3

The table below includes a partial list of genetic syndromes associated with pheochromocytomas (and/or paragangliomas, generally):

Disease	Gene	Associated Lesions
MEN 2A (Sipple syndrome)	*RET*	Pheochromocytoma Medullary thyroid carcinoma Parathyroid hyperplasia
MEN 2B (Wagenmann–Froboese syndrome)	*RET*	Pheochromocytoma Medullary thyroid carcinoma Mucocutaneous ganglioneuromas
Neurofibromatosis, type 1 (von Recklinghausen syndrome)	*NF1*	Pheochromocytoma Café-au-lait spots Optic nerve glioma
von Hippel–Lindau syndrome	*VHL*	Pheochromocytoma Renal cell carcinoma, renal cysts Hemangioblastoma Pancreatic neuroendocrine tumors Retinal angioma
Hereditary paraganglioma– pheochromocytoma syndrome	*SDHB*, *SDHD*	Pheochromocytoma Paraganglioma

Pheochromocytoma and paraganglioma are more commonly associated with an inherited mutation than any other cancer.

This patient's pheochromocytoma thus likely has a genetic underpinning—and that's quite common. In fact, pheochromocytoma is the most heritable tumor in the sense that the largest proportion of cases can be explained by inherited genetic mutations. Several associated disorders are described above, and this is only a subset of the various syndromes that contain pheochromocytoma as an association. While the eponyms have gone out of style for most of these syndromes (MEN2A, MEN2B, and neurofibromatosis, type 1 [NF-1], at least), they're still listed above for completeness's sake. The gene with the associated pathological mutation is listed, as are several associated lesions. It's worth learning these associations, as they're important for screening in patients who are diagnosed with one of these syndromes.

PHEOCHROMOCYTOMA: SUMMARY

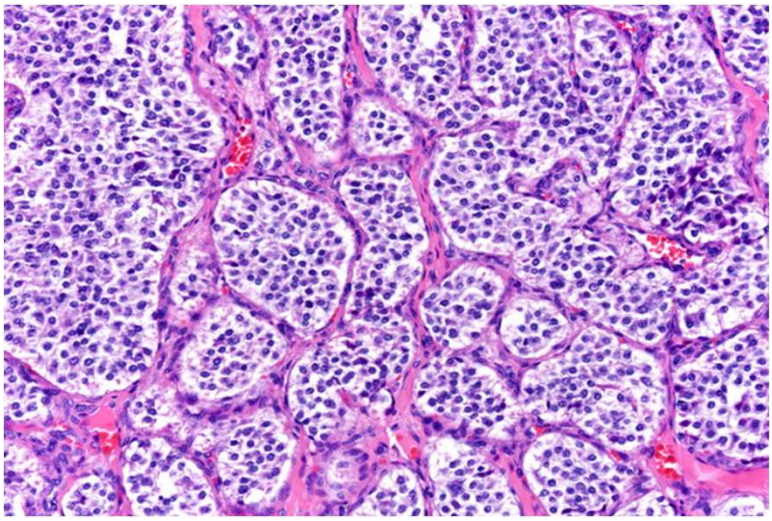
The patient does well after surgery and is referred to a genetic counselor for screening and further guidance.

Let's review key features of the histopathology of this tumor.



gross appearance

well-circumscribed tan-brown tumor, often with hemorrhage or necrosis



histopathology

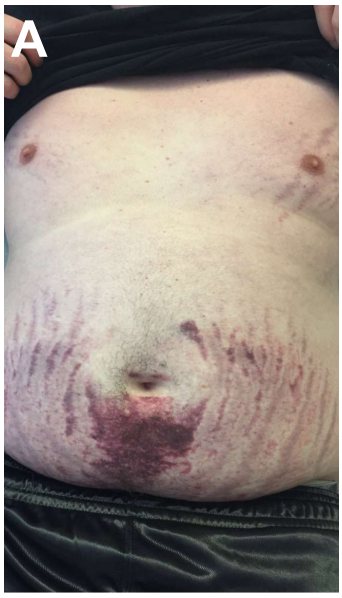
"zellballen" (nests of tumor cells), granular and basophilic cytoplasm, delicate vessels

As a brief review, now that our patient is recovering: **pheochromocytomas** are usually benign (90%) tumors that are well-circumscribed with pushing borders and have a tan-brown color. They may have hemorrhage or necrosis, but those are not signs of malignancy in this tumor type. And microscopically, they contain nests of cells with granular basophilic cytoplasm (often referred to as zellballen) separated by fibrous septa containing delicate vessels. The example shown here has more prominent septa than the one shown previously.

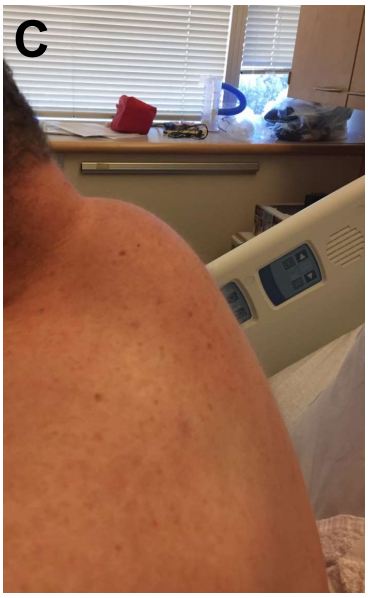
Case #4

CASE #4

A 38-year-old patient presents with complaints of a 20-pound weight gain, fatigue, and facial puffiness over the past six months. They report that they maintain a healthy diet and regular exercise routine, but have still gained significant weight, especially around the abdomen and face. They have muscle weakness and note changes in their skin, including easy bruising and new purple, linear marks. Physical examination shows the following:







In our fourth and final case for this TBL session, we're presented with a young to middle-aged patient with weight gain (despite a self-described healthy diet and regular exercise routine), fatigue, and facial puffiness, muscle weakness, and skin changes as shown in the physical examination findings. These physical characteristics are classic for the diagnosis of interest in this case and will be described further on the next slide.

CASE #4

Q1: What is shown in each image? What does each indicate?

A: violaceous abdominal striae (stretch marks), indicating thinning of the skin and reduced structural integrity of its connective tissue.
B: "moon facies," indicating lipodystrophy, or redistribution of the subcutaneous fat to the facial area.
C: "buffalo hump," another indication of lipodystrophy, or redistribution of the subcutaneous fat to the dorsocervical region.

Q2: What diagnosis tops your differential diagnosis? How might you confirm this diagnosis?

Cushing syndrome (hypercortisolism) is the most likely diagnosis. We do not know what is driving this hypercortisolism; thus, an appropriate workup may include:

- Additional history (other symptoms of Cushing include hirsutism [hair growth], menstrual irregularities, diabetes, and psychological manifestations like depression or irritability)
- Medication list (exogenous steroids)
- Endocrine testing (plasma cortisol, 24-hour urine free cortisol, plasma ACTH concentration, response to dexamethasone suppression)
- Imaging (to be discussed later)

The physical examination shows several findings suggestive of **Cushing syndrome** (**hypercortisolism**). More specifically, in the first image (A), we see violaceous (purple) abdominal **striae**, or stretch marks. These may be seen in several different disorders, but the number and distribution of them (on the abdomen and, often, the thighs) is suggestive of Cushing syndrome. The underlying pathophysiology is a reduction in the structural integrity of the connective tissue of the skin (predominantly in the dermis) that leads to scarring. In the second image (B), we see so-called "**moon facies**," which represents a form of **lipodystrophy**, or the redistribution of subcutaneous fat. Moon facies is due to that redistribution ending up in the facial area, especially in the cheeks. Finally, in the third image (C), we see that same redistribution to the back of the neck, forming a "**buffalo hump**."

That physical examination should prompt us to consider causes of Cushing syndrome. Most common in the United States would be the used of exogenous, prescribed corticosteroids. The patient's medical history and medication list should be reviewed. We should take stock of any additional symptoms of Cushing syndrome, as well: hirsutism, menstrual irregularities (if relevant), diabetes mellitus, and psychological manifestations like mood disorders.

If the medical history and medication list are unrevealing, we are likely looking at a case of Cushing syndrome due to endogenous corticosteroid production. Several endocrine tests will be warranted, as will be described on the next few slides. Imaging is also warranted to look for a mass to explain these symptoms—but of which anatomic region(s)?

CASE #4

The labs return as follows:

Test	Value	Normal Range
Plasma glucose	300 mg/dL (**H**)	70–110 mg/dL
24-hr urine free cortisol	200 mcg/24 hr (**H**)	<50 mcg/24 hr

And based on these results, the patient is admitted to the hospital for further testing:

Test	Value	Normal Range
Low-dose dexamethasone suppression test (serum cortisol, 8 am)	12 mcg/dL (**H**)	<5 mcg/dL
Plasma ACTH (late afternoon)	125 pg/mL (**H**)	<20 pg/mL
High-dose dexamethasone suppression test (serum cortisol, 8 am)	2 mcg/dL	>50% cortisol reduction after administration

We'll look at the recommended algorithm for the workup of Cushing syndrome on the next slide, but let's make some sense of these initial tests before revealing it. The central endocrinopathy of Cushing syndrome is of hypercortisolism. What does cortisol do? Well, for one, it is a **counterregulatory hormone** of insulin, so it will raise blood glucose levels. Therefore, **plasma glucose** and **24-hour urine free cortisol** are sensible as initial tests. They should both be elevated in Cushing syndrome.

Then, we may consider a **low-dose dexamethasone suppression test**. With normal physiology, dexamethasone (a potent corticosteroid) should cause negative feedback in the central nervous system and lead to a drop of corticotropin-releasing hormone (from the hypothalamus) and adrenocorticotrophic hormone (from the anterior pituitary). As such, endogenous cortisol levels should drop. This test may be performed overnight, with cortisol levels measured in the morning when they usually peak. In Cushing syndrome, this feedback mechanism will fail to suppress the production of cortisol, and levels will remain elevated.

But is the problem at the level of the adrenal, or of the anterior pituitary? ACTH can help us figure that out. If it is elevated, then the hypercortisolism is presumably ACTH-dependent and the pathology is within the anterior pituitary (or, less commonly, the hypothalamus). If it is low, then the hypercortisolism is independent of ACTH production and is due to pathology within the adrenal cortex.

There's another possibility, too: the ACTH may be coming from the anterior pituitary, or it may be coming from an ectopic source (like small cell lung carcinoma, which may produce it as a paraneoplastic phenomenon). To distinguish these possibilities, a **high-dose dexamethasone suppression test** may be performed. While the low-dose test is insufficient to suppress ACTH production by the anterior pituitary, the high-dose test is generally successful at suppressing endogenous ACTH (and therefore cortisol) production. Ectopic production of ACTH couldn't care less about the negative feedback loop and will demonstrate elevated cortisol levels, even with the high-dose dexamethasone suppression test.

CASE #4

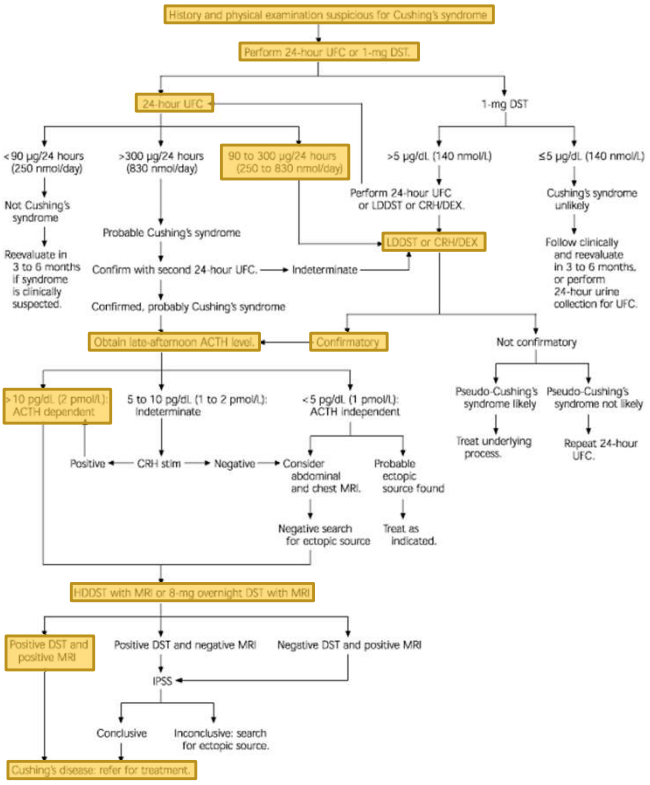
Q3: Do the patient's labs support the diagnosis of Cushing syndrome? How about Cushing disease?

The 24-hour urine free cortisol confirms hypercortisolism (Cushing *syndrome*), so we move on to a low-dose dexamethasone suppression test (DST).

In that test, 1 mg dexamethasone is given at night and the cortisol measured the next morning. The result was an elevation of cortisol, a confirmatory low-dose DST.

ACTH is measured and is found to be high; the hypercortisolism is ACTH-dependent.

Then, a high-dose DST is given. The cortisol levels drop by >50% the next morning. This is seen when the tumor responds to negative feedback (arising from the pituitary, Cushing *disease*) as opposed to ectopic sources (where cortisol levels will fail to suppress).



This flowchart basically describes what was explained on the previous slide. The algorithm looks confusing at first glance, but the logic of it should make quite a bit of sense. In our scenario, the patient had 24-hour urine cortisol, a low-dose dexamethasone suppression test that failed to suppress cortisol production, elevated ACTH, and a high-dose dexamethasone suppression test that *did* suppress cortisol production. These findings, in conjunction, suggest that the findings are driven by ACTH production in the anterior pituitary.

This question also asks us to distinguish between Cushing syndrome and Cushing disease. Let's make sure we're clear on the terminology. **Cushing syndrome** simply describes the state of hypercortisolism, with its associated symptoms (as described in this patient's physical examination and elsewhere in this case). It may be due to many causes, but the common feature is of high cortisol. Of note, the term Cushing syndrome may also be applied to iatrogenic (healthcare-caused) cases, when exogenous steroids are given—even if the elevated hormone is not technically cortisol in those cases. A subset of Cushing syndrome is due to **Cushing disease**: ACTH-dependent hypercortisolism due to an underlying ACTH-producing pituitary tumor.

CASE #4

Q4: How would serum ACTH, the response to a high-dose dexamethasone suppression test, and the gross appearance of the adrenal gland appear in (1) a pituitary adenoma; (2) a case of ectopic ACTH production, such as by small cell lung carcinoma; and (3) an adrenal cortical neoplasm?

Etiology of Cushing syndrome	ACTH	High-dose DST	Gross appearance
Pituitary adenoma (~70% of cases due to endogenous factors*)	↑	Suppressed	Bilateral cortical hyperplasia
Ectopic ACTH production (i.e., small cell lung carcinoma; ~10%)	↑↑	Not suppressed	Bilateral cortical hyperplasia
Adrenal cortical neoplasm (adenoma ~10%; carcinoma ~5%)	↓↓	Not suppressed	Unilateral cortical nodule

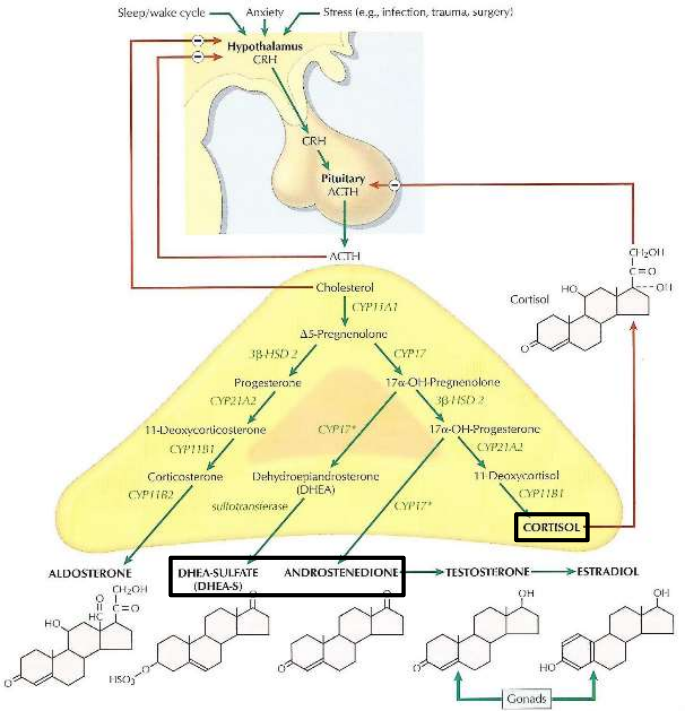
* NOTE: Percentages refer to the frequency of *endogenous* Cushing syndrome; most cases are actually due to exogenous administration of glucocorticoids (i.e., iatrogenic Cushing syndrome)

This table shows how three common causes of Cushing syndrome (hypercortisolism) would play out with respect to these common tests. Let's briefly consider each:

- With a pituitary adenoma, the hypercortisolism is driven by the excess production of ACTH. Thus, ACTH levels will be high. But the anterior pituitary is usually responsive to suppression by a high dose of dexamethasone, so the high-dose DST should show that suppression. And given that the hypercortisolism is driven by ACTH, the adrenal cortex will become hyperplastic in response to this increased trophic signaling, manifesting as bilateral cortical hyperplasia.
- With ectopic ACTH production, usually a paraneoplastic phenomenon, a tumor like small cell lung carcinoma produces the ACTH. The tests will look similar to a pituitary adenoma with two notable differences: ACTH is usually *markedly* elevated (more so than with a pituitary adenoma) and the high-dose DST will fail to suppress the production of endogenous steroids.
- Finally, with an adrenal cortical adenoma or carcinoma that produces cortisol, negative feedback will cause ACTH levels to be markedly depressed. The high-dose DST, which normally should suppress ACTH production in cases of hypercortisolism that are ACTH-dependent, will have no effect here: the hypercortisolism in this case is ACTH-*independent*. Usually, we'll see a unilateral nodule, like we saw in Case #1 of this TBL session.

CASE #4

Q5: What is the biochemical basis for the signs and symptoms seen in Cushing syndrome?



We focus mostly on the role of cortisol in Cushing syndrome:

- Increases hepatic gluconeogenesis and decreases peripheral glucose uptake (by downregulating GLUT-4): leads to **diabetes mellitus/impaired glucose tolerance**
- Increases proteolysis, decreases collagen synthesis, and decreases osteoblastic activity: leads to **muscle wasting**, **thinning of skin** with **bruising** and **striae**, **osteoporosis**
- Diurnal (24-hour) variation in levels is lost: leads to **psychiatric disturbances** (commonly **depression**)
- Some mineralocorticoid activity of cortisol: leads to **hypertension**

But there are also effects of DHEA-S and androstenedione, which are converted peripherally to testosterone: **acne**, **hirsutism**, and **menstrual irregularities** (if relevant)

Aldosterone is *usually* not increased in Cushing syndrome.

The adrenal cortex produces several hormones, including cortisol, aldosterone, dehydroepiandrosterone-sulfate (DHEA-S), and androstenedione. Of those, the one we tend to focus on the most in Cushing syndrome is cortisol. As a steroid hormone, cortisol mediates changes in transcription to lead to its effects. This includes the specific pathway effects described on the slide, thematically organized around: glucose regulation, connective tissue synthesis, psychiatric components, and (weak) mineralocorticoid activity. The sex hormones, DHEA-S and androstenedione also lead to hormonal effects, including acne, hirsutism (hair growth), and menstrual irregularities (in those with uteri).

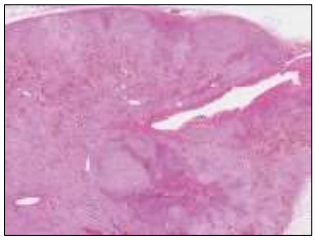
Of note, aldosterone is usually not increased significantly in Cushing syndrome. This is because the regulation of its pathway is predominantly through the renin–angiotensin–aldosterone system, rather than ACTH within the hypothalamus–pituitary–adrenal (HPA) axis.

CASE #4

This patient would *not* undergo a biopsy or resection, but as the endocrine pathologist on service, you consider what the patient's adrenal glands might look like.



Adrenal VS 4-1



Adrenal VS 4-2

Take five to ten minutes to examine the two virtual slides and answer the questions on the following three PowerPoint slides.

CASE #4

Examine Adrenal VS 4-1 and answer the following questions:

Q6: Given the likely etiology of the patient's hypercortisolism, could their adrenal glands look like the ones in the black (solid) box? If not, could ANY patient whose glands look like this have hypercortisolism?

These adrenal glands are not consistent with our patient's likely etiology but could be seen with chronic use of exogenous steroids. Those steroids would suppress the endogenous hypothalamus-pituitary-adrenal (HPA) axis and lead to atrophy of the gland.

Q7: Given the likely etiology of the patient's hypercortisolism, could their adrenal glands look like the ones in the yellow (dashed) box? If not, could ANY patient whose glands look like this have hypercortisolism?

This is likely the appearance of our patient's adrenal glands: they are hyperplastic (enlarged compared to normal, due to an increased number of cells). This hyperplasia is driven by the excess of ACTH signaling from the anterior pituitary.

Of note, the appearance of the adrenal glands could be highly variable and still result in the same finding of Cushing syndrome. Our patient's glands are most likely hyperplastic, having been stimulated by excessive production of ACTH by the anterior pituitary; this (Cushing disease) is the most common *endogenous* cause of Cushing syndrome in the United States (the most common cause overall is iatrogenic Cushing syndrome cause by chronic administration of steroids). That excessive ACTH leads to hyperplasia of adrenal cortex, as seen in the bottom glands in the virtual slide.

But the adrenal glands could look just the opposite and still lead to the same clinical findings. An example for this would be adrenal cortical suppression due to long-term use of exogenous steroids. These exogenous steroids would suppress endogenous ACTH production, potentially leading to atrophy of the patient's glands. This is why it is important to taper steroids, if given chronically. The patient's own glands would need time to turn back on and function.

CASE #4

Examine Adrenal VS 4-2 and answer the following questions:

Q8: Start with a view at low power. How would you describe the overall appearance of this gland, histologically? Pay particular attention to the vague structures indicated by the black dotted-line circles.

The gland is clearly quite enlarged and hyperplastic. The thickness of the gland overall is increased. There is evidence of increased blood flow throughout the gland, given how numerous the cross-sections of red-staining vessels there are appreciable even at low power. The dotted circles show vague nodularity throughout the adrenal cortex without a single, discrete mass. The appearance of the patient's adrenals on both sides would be similar.

Q9: What is the tissue indicated by the green arrow?

This is the capsule of the adrenal gland, composed of dense irregular connective tissue.

On histology, we can appreciate that the adrenal cortex is diffusely hyperplastic. It is hard to find any recognizable adrenal medulla on this slide. The thickness of the gland is markedly increased compared to normal, and this increase is predominantly due to hyperplasia of the zona fasciculata, as described in the next PowerPoint slide. Endocrine tissues, generally, are quite vascular because they need to dump the hormones they produce into the circulation. Here, we can appreciate prominence of this circulation, as there are numerous small capillaries visible on the slide. Finally, looking at the parenchyma, the hyperplasia has a vaguely nodular pattern. The “nodules” are poorly circumscribed and unencapsulated, but there’s general sense of some large-scale spherical structure to the hyperplasia. Given the systemic nature of the high ACTH driving this patient’s pathology, the adrenals would show this same diffuse, nodular hyperplasia bilaterally.

We also note here that the adrenal gland is surrounded by a relatively thin capsule of dense irregular connective tissue, just because we didn’t acknowledge it in any of the previous cases.

CASE #4

Continue to examine Adrenal VS 4-2 and answer the following question:

Q10: Examine the layers indicated by 1, 2, and 3 in the blue polygon. What are these layers? Which layer(s) demonstrate the most hyperplasia compared to normal?

1: zona glomerulosa (produces mineralocorticoids, like aldosterone)
2: zona fasciculata (produces glucocorticoids, like cortisol)
3: zona reticularis (produces sex hormones, like DHEA-S and androstenedione)

Layer 2, the zona fasciculata, is clearly markedly hyperplastic. It not only is much wider than normal, but also has started to take on a nodular appearance, as described in the previous questions and is growing into the other two zones. The zona glomerulosa and zona reticularis are rendered discontinuous in the specimen because of this ingrowth from the zona fasciculata. This correlates with the patient's hypercortisolemia. In some cases, although not particularly evident in this case, the zona reticularis may also be hyperplastic. Hyperplasia of the zona glomerulosa is less common because its regulation is more strongly performed by the renin-angiotensin-aldosterone system than the ACTH pathway.

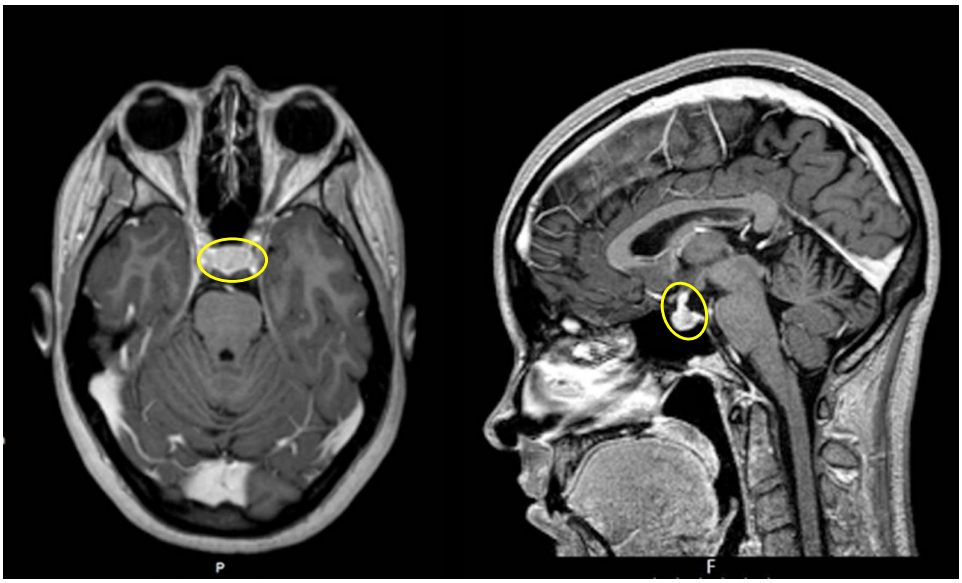
The normal architecture of the adrenal cortex is *somewhat* maintained but is certainly a bit disturbed. The adrenal cortex normally has three layers, which can be observed on this virtual slide:

- The most superficial **zona glomerulosa** produces mineralocorticoids, like aldosterone. It is generally composed of small clusters of well-defined cells with scant to moderate cytoplasm.
- The middle **zona fasciculata** produces glucocorticoids, like cortisol. It is generally composed of cords and bands of cells, usually two cells wide, with vacuolated cytoplasm.
- The deepest **zona reticularis** produces sex hormones, like DHEA-S and androstenedione. It is generally composed of haphazardly-arranged, moderately-sized cells with granular, eosinophilic cytoplasm.

The zona fasciculata here is markedly hyperplastic and starts to interrupt the bands corresponding to the zona glomerulosa and zona reticularis. The zona reticularis may also be slightly hyperplastic, although not particularly significantly in this specimen. (The zona reticularis usually accounts for <10% of the thickness of the cortex, so it is likely a bit thicker than normal here.) The zona glomerulosa is not notably hyperplastic, because while ACTH does weakly drive aldosterone secretion, the stronger regulator of aldosterone production is angiotensin II.

CASE #4

Based on the patient's lab testing, an MRI of the head was performed:



Q11: What is shown in the imaging? What is the final diagnosis?

This is a pituitary adenoma, which is presumably secreting ACTH. That makes the final diagnosis in this case Cushing disease: Cushing syndrome (hypercortisolism) driven by an ACTH-secreting tumor in the anterior pituitary.

This patient's MRI reveals what we should have been expecting: a **pituitary adenoma**, presumably secreting ACTH to cause their findings. Diagnostically, this is called **Cushing disease**: that is, Cushing syndrome due to an ACTH-producing pituitary tumor.

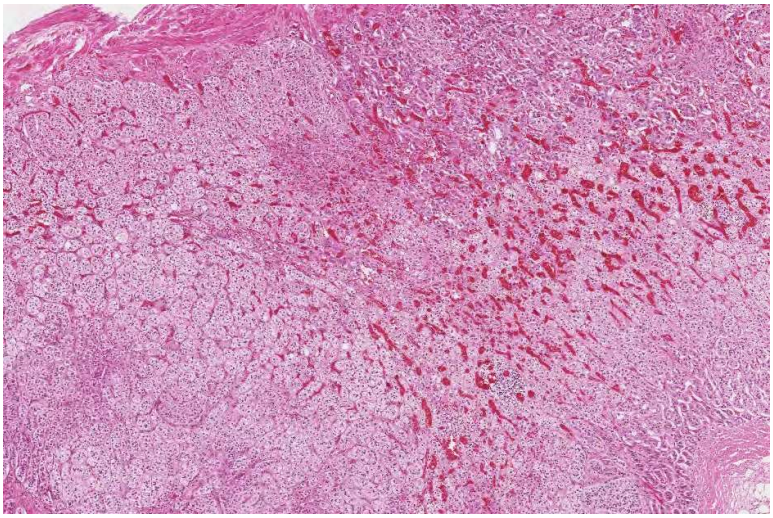
CUSHING SYNDROME AND DISEASE: SUMMARY

The patient is scheduled for a transsphenoidal resection of the pituitary adenoma, which is successful. Their symptoms slowly improve.

Let's review key features of the histopathology of this lesion, when due to Cushing disease (most common endogenous cause in the U.S.).



gross appearance
(bilaterally) hyperplastic adrenal glands, affecting the adrenal cortex



histopathology
varying degrees of hyperplastic change, particularly affecting zona fasciculata ± reticularis

Alright, we've completed our final case. Our patient was successfully treated by resecting the ACTH-producing adrenal adenoma. Remember from head and neck anatomy that a transsphenoidal approach would be the best way to reach the sella turcica (going through the nasal cavity, nasopharynx, and sphenoid sinus behind them to reach the pituitary gland).

Most of this case focused again on the physiology and pathophysiology of the hypothalamic–pituitary–adrenal (HPA) axis and less on the gross and microscopic pathology for the same two reasons as were seen in Case #2. One, if we know the mechanism of the disease (ACTH-driven hypercortisolism from the adrenal cortex), the gross findings (thickening of the cortex) and histopathology (hyperplasia of the zona fasciculata) flow very naturally from that understanding. Second, this isn't a disease that we diagnose with a biopsy or resection, unless the patient died acutely, and the diagnosis is made postmortem in an autopsy.

LEARNING OBJECTIVES

By exam time, you should be able to do the following:

- Correlate the gross and histopathological features of a pheochromocytoma with its tissue of origin and its clinical manifestations.
- Describe and identify genetic syndromes commonly associated with pheochromocytomas.
- Correlate the gross and histopathological features of an aldosterone-producing adrenal cortical adenoma with the clinical findings of Conn syndrome.
- Correlate the gross and histopathological features of adrenal atrophy (for example, due to autoimmune adrenalitis) with the clinical findings of Addison disease.
- Describe additional causes of acute adrenal insufficiency, including Waterhouse–Friderichsen syndrome, hypopituitarism, and steroid withdrawal.
- Correlate the gross and histopathological features of secondary adrenal hyperplasia with the clinical findings of Cushing syndrome or disease.

The Learning Objectives (LOs) are included here. Note that the order of the LOs does not match the order of the cases in the TBL; this is by design. It encourages you to think through each case, rather that simply matching it to its LO to determine the diagnosis of interest. Ideally, you'll be looking at these notes *after* the TBL has concluded, during review, so here is the association of each LO with the case that represents it:

• Correlate the gross and histopathological features of a pheochromocytoma with its tissue of origin and its clinical manifestations. **Case 3**
• Describe and identify genetic syndromes commonly associated with pheochromocytomas. **Case 3**
• Correlate the gross and histopathological features of an aldosterone-producing adrenal cortical adenoma with the clinical findings of Conn syndrome. **Case 1**
• Correlate the gross and histopathological features of adrenal atrophy (for example, due to autoimmune adrenalitis) with the clinical findings of Addison disease. **Case 2**
• Describe additional causes of acute adrenal insufficiency, including Waterhouse–Friderichsen syndrome, hypopituitarism, and steroid withdrawal. **Case 2**
• Correlate the gross and histopathological features of secondary adrenal hyperplasia with the clinical findings of Cushing syndrome (disease). **Case 4**

Adrenal Pathology



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Image: artsy interpretation of the adrenal glands secreting hormones (although it would do so by releasing them into the blood stream, not through paracrine release as is implied in this image)